



ATAG

Atag Celsius induction cooker

Phase 3 | Redesign Proposal

Team 8

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Preface

The aim of this project is to enhance the usability and user experience of the Atag Induction Celsius Cooking hob. This new advanced cooking hob allows users to cook with precise temperature control. The current hob on the market features three different cooking zones, timer, and stopwatch functions, and offers a new cooking experience that includes slow cooking to complement the ingredients.

The hob is operated through a graphical interface, and users can enhance their interaction with the Celsius accessories: a probe and a Celsius cooking pan, which provide even more precise temperature control. The goal of the Atag Celsius cooking hob is to provide users with the tools to cook like a professional chef at home, offering control and a comprehensive overview of the cooking process.

The team



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Introduction

In this report, we will delve into the comprehensive usability analysis for the Celsius Cooking hob, which is the culmination of previous insights and aims to present a detailed redesign of the interface focused on cooking with temperatures (see Figure 1).

Our goal is to propose a redesign that minimizes changes to the current technology while maximizing improvements in the user flow, keeping the positioning of the induction hob the same to maintain Atag's market position. The proposed improvements have been informed by user testing observations from Phase one and tested design directions from Phase two, focusing on enhancing the user experience within the temperature zone.

The ultimate design goal is to create a new, intuitive interface for the Atag Iris Slider, empowering regular cooks to feel in control and excited during the cooking process. The report will outline the final refinement of the design, and the decision process leading to these enhancements, and wrap up with design detailing, recommendations, and advice for implementation based on new user testing and usability metrics.

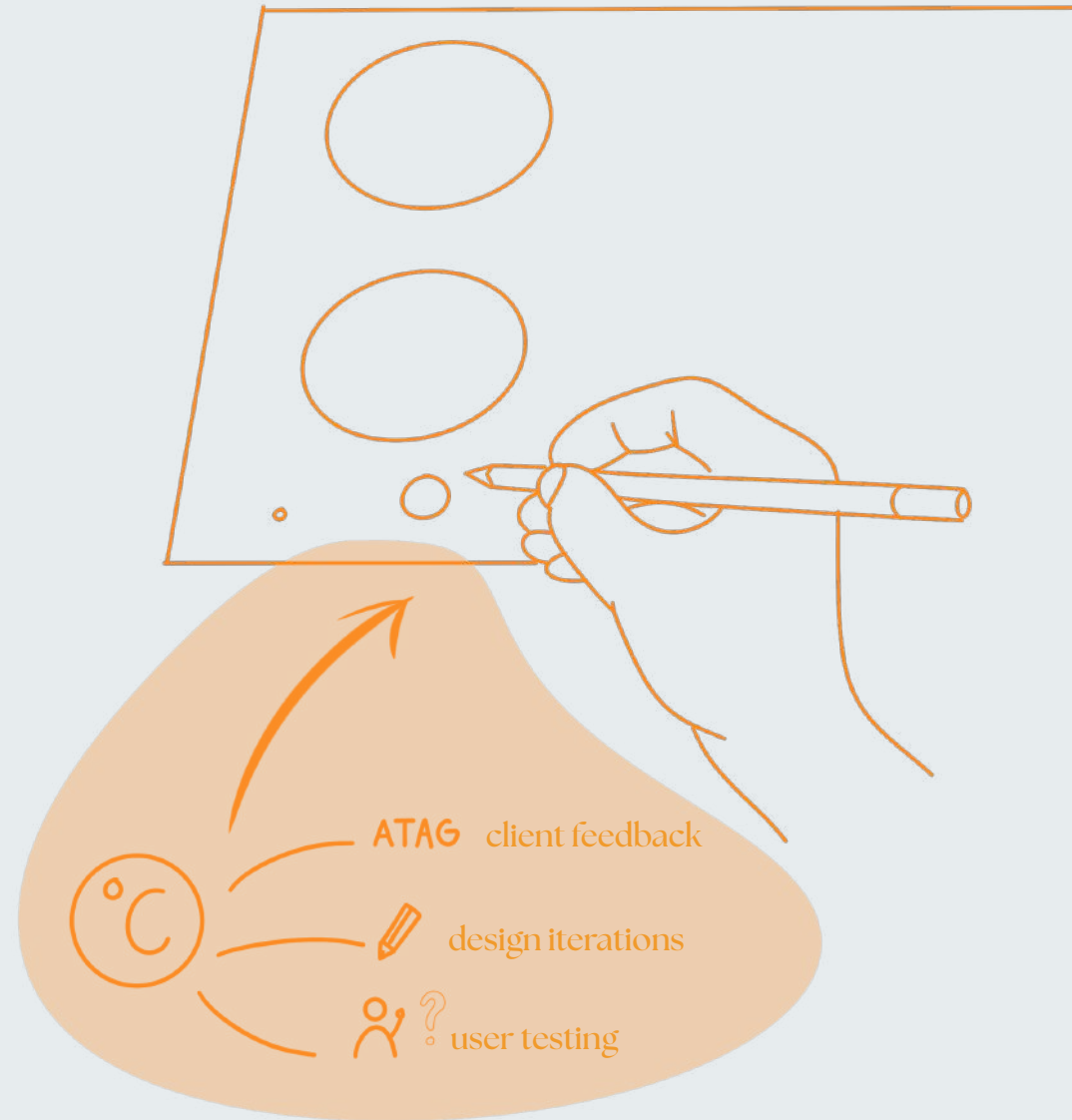


Figure 1: How we came to the design visualised.

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1 ■ Recap of Phase 2

This section presents an outline of the previous conceptualising Phase of our project, primarily focusing on the conclusions derived from it. It commences by summarizing our design choices and ends with our initial proposal for redesign.

Quick Chapter glance:
Design Decisio



Design decisions

In Phase 2, several design decisions were made for the first redesign proposal. This chapter shows a quick recap of the most important changes up until the start of Phase 3.

Conscious choice

When the hob is switched on, users have different cooking modes to choose from (see Figure 2). This enables them to quickly start cooking or provide a simpler display for guests. This empowers users to feel more in control of their cooking style. Additionally, the Celsius cooking icon has been refined and the settings are readily accessible once the hob is on. To make the lock function of the hob more distinct, it has been separated from the on/off button. The lock turns orange when turned on, and is grey when not active.

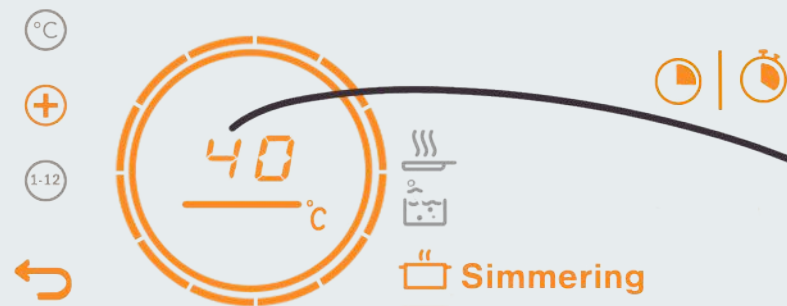


Figure 3: Visual of interface.

Presets

The display has been improved to make cooking more intuitive. It now provides users with information on various presets, including text next to each icon to help them understand which mode they're in as presented in Figure 3. If you swipe to change the temperature in increments of 5 degrees and go past the start point of another icon, the icon will automatically change to match your intended setting.

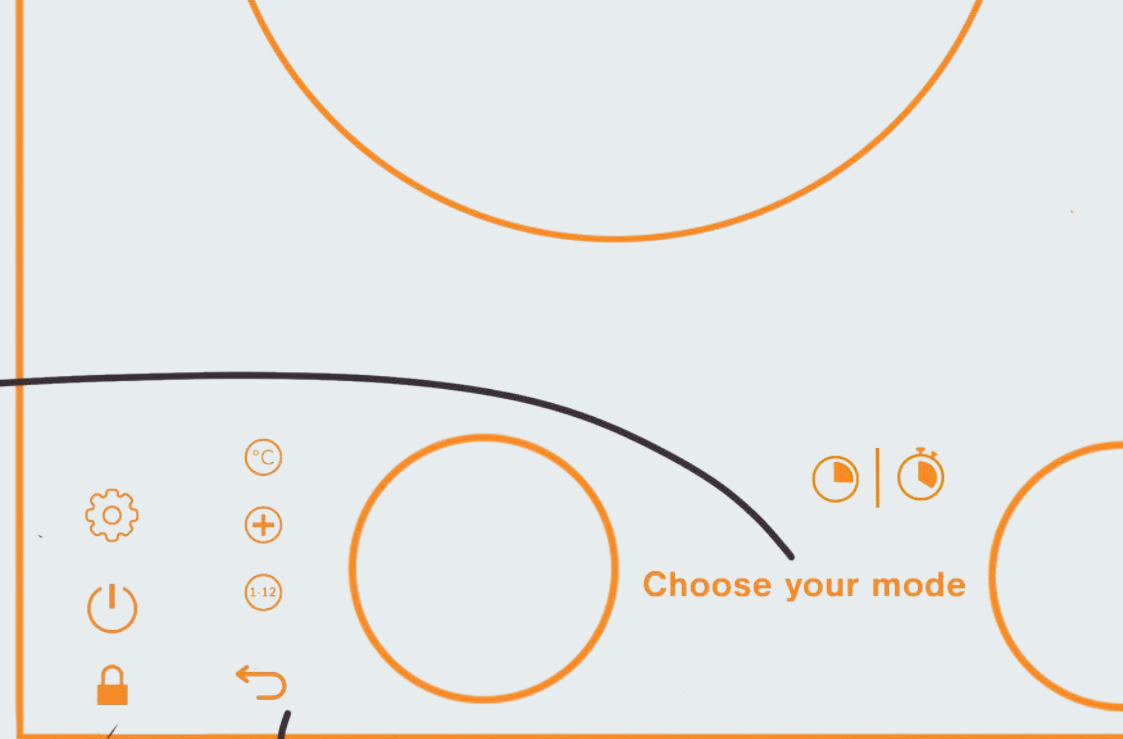


Figure 2: Visual of initial redesign on the interface.

Consistent presets

Once you've selected your mode, it will always be displayed on the left of the IRIS slider, as shown in Figure 2. The Celsius and plus cooking modes use the same icons and default temperatures. Additionally, the back button has been added to the display, making it easy to navigate through modes.

Default temperatures presets

The temperature presets are configured in an improved way. Instead of providing a fixed set of temperatures, the preset now serves as a starting point for temperature adjustment. The user can increase or decrease the temperatures in increments of 5 degrees from the default temperatures. This gives them a greater sense of control and avoids being limited to a predetermined range of temperatures.

Design decisions

Preheat

The preheat function has undergone significant changes to make it more user-friendly and intuitive, see Figure 4 for the visualisation of the interface. To address that the user is 'locked' and has to wait, the back button is no longer displayed. Additionally, instead of the countdown, which can be stressful, a filling heating sign is now displayed beneath the temperature to guide the user in the process. To indicate that the hob is in preheat mode, this is displayed above the mode they are currently in. Also, to show that the pan is boosting, which is the reason why the user is locked for a short time, the hob displays the word "boost", keeping the user informed throughout the process.

Timer function

Finally, in our effort to improve the cooking experience, we have enhanced the timer function to make it more intuitive and easy to use. The timer works the same way as before, but now a small grey bar is displayed within the IRIS slider to show the progress of time as shown in Figure 4. To reduce cognitive load and avoid stress while cooking, the numbers representing the time disappear and only appear after the timer function is pressed. This means that the primary way to check the remaining time is through the grey bar. This feature allows for a quick glance at the timer's progress and helps users feel excited instead of pressured while cooking.

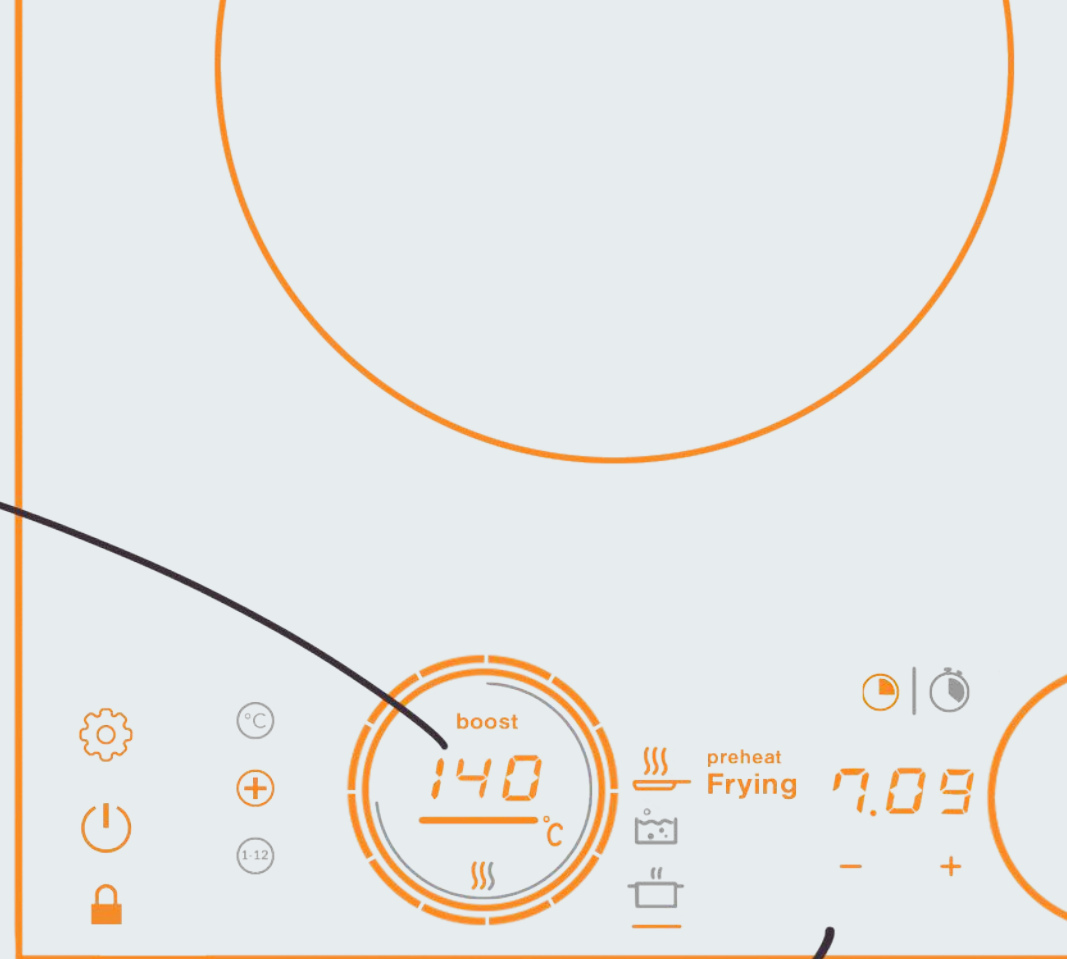


Figure 4: Visual of interface.

2. Design refinement

In this chapter, we move from the initial redesign proposal to a more detailed plan. We begin by gathering feedback on the ideas we have presented, which will serve as the basis for redefining our concept as we move into the detailed redesign Phase. The feedback included input from peers in the IDE program and feedback from Atag during a client meeting. This input enabled us to create small prototypes to reinforce our concept. This chapter provides an overview of the feedback received and the weaknesses in the design that still need to be validated through rapid small-scale testing. Finally, we present the final design refinement before moving on to user testing of our redesign.

Quick Chapter glance:

Client Feedback

Peer Feedback

Small prototyping

Conclusions



Client Feedback

After sending the 'Design Brief' from Phase 2's report to our client, we held a meeting with them to gain valuable feedback. The meeting with Björn Flinterhoff and Vincent Hofstee was positive overall. However there

were 3 important discussion points we would like to mention. It allowed us to reconsider some aspects of the redesign in a more practical and feasible way. They were:

The Timer

A part of the design that they we were discussed with us was the grey timer circle inside of the Iris slider as shown in Figure 5.



Figure 5: Visual of the timer.

We have to consider the display of the circle for long hours (e.g., when slow cooking for 5 hours). The visual countdown and its approach to completion may be needed to be made more explicit in the next iterations.

The Touchscreen

In our redesign the display was intended as touchscreen (Figure 6), where text would be revealed as the icons were pressed. The clients mentioned that touchscreen would come with extra



Figure 6: The current redesign touchscreen.

cost .

So as a solution to this, we intend to redesign keeping in mind the creation of a graphical interface with no overlapping text and each icon having its own place.

The Text

In our redesign we included textual elements, as shown in Figure 7. However, we learned to consider that some markets do not allow any English. While the word 'Boost' is permitted, Atag



Figure 7: The current redesign with text.

has aimed to minimize written text as much as possible. This is because they may need to provide translations for international markets like China, as well as for local markets. It's important to think about the effectiveness and ease of understanding of visuals versus text.

Peer evaluation

At the start of Phase 3, a peer evaluation session was held. Together with other teams in the same studio, we divided ourselves up into 5 groups and explained the product and our redesign proposal to other students. Based on this, the other students gave us feedback and suggestions on our redesign. This evaluation served as a first impression of the new interface and it provided us with valuable information on the intuitivity and shortcomings of the redesign. The feedback and suggested improvements we collected from this session are visible in Figures 8 & 9.

Feedback

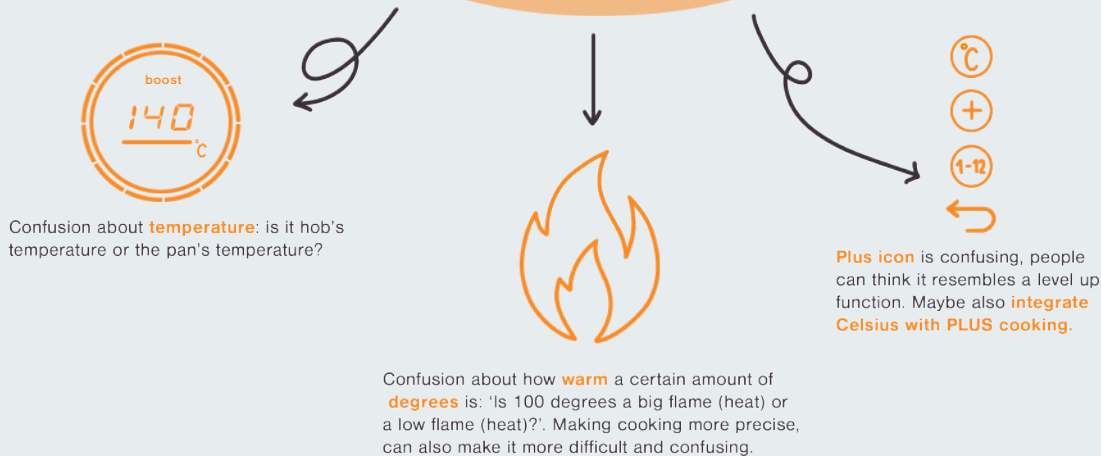


Figure 8: Overview of the feedback of the peer evaluation.

Suggestions

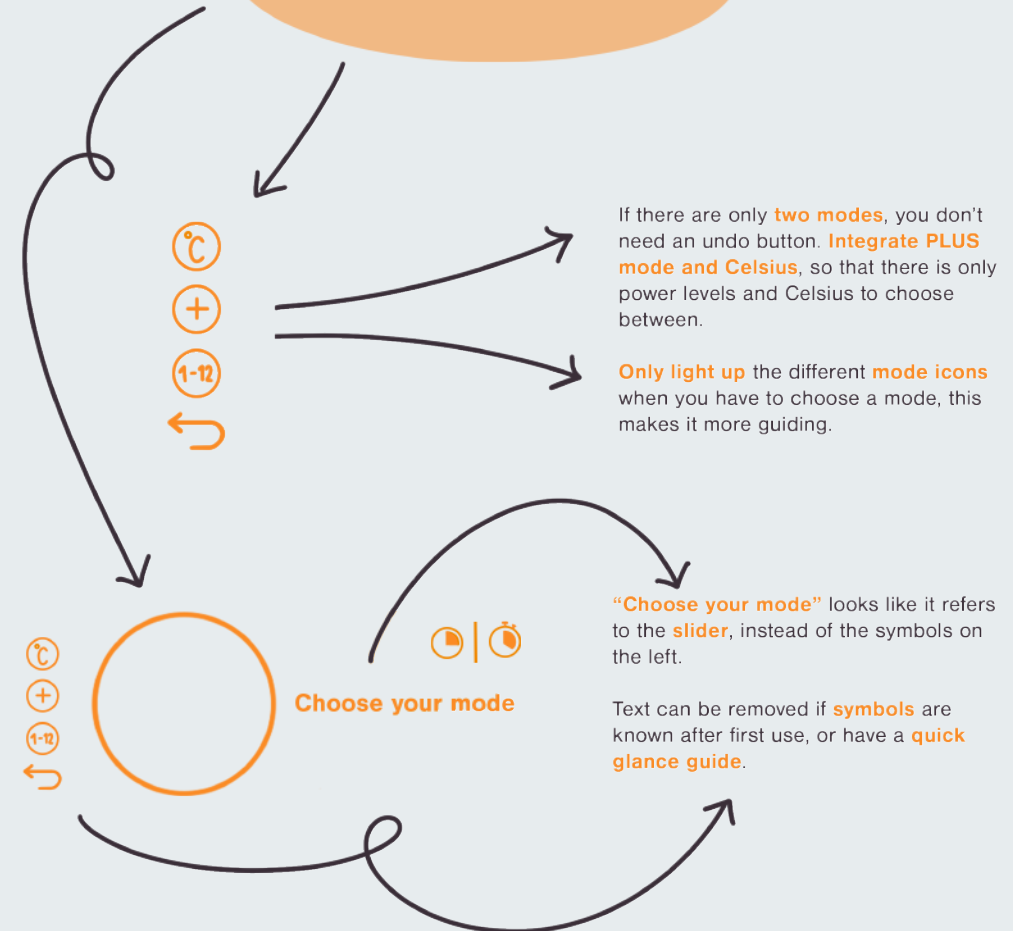


Figure 9: Overview of the suggestions made according to the peer evaluation.

Small prototyping

Before examining the final user test, the client and peer feedback was implemented in the redesign. This is done in the following manner. From the feedback, six improvement areas belonging research question were derived. For each research question a small test was done among less than 10 people supported by Lofi prototypes. Below this text, the improvement areas are presented. A small test plan was made beforehand, so every team member had the same structure for testing to get useful results.

Improvement areas

Placement iris slider

What is the best placement of the Iris slider?

Symbols and text

Are the symbols clear in what they mean and is the text neglectable?

Timer function

How can the timer function be used if used for really long during slow cooking?

Celsius mode

Can celsius be intergrated within the PLUS cooking? So the user only has two modes?

Heat indication

How is it best communicated to show how 'hot' the celsius is? Is this valuable to connect with the powerlevels?

Temperature increment

What is the best temperature increments setting?

Placement Iris slider

What is the best placement for the Iris slider?

Why

During the peer evaluation the following feedback was given: “Choose your mode” feels like it is referring to the empty iris slider instead of the lines. The following idea was derived to change the layout a bit, so the text is next to the buttons it referred to. To explore if this is indeed a better option, the choice was made to do a small user test with both of the layout options.

How

For the first test, two placements of the Iris slider were made. (See Figure 10) These layouts were printed out. Seven people within the faculty of IDE were asked to look at the different layouts and give their opinions about which layout they thought was more intuitive and more logical. Especially important for them was to explain why they thought so.

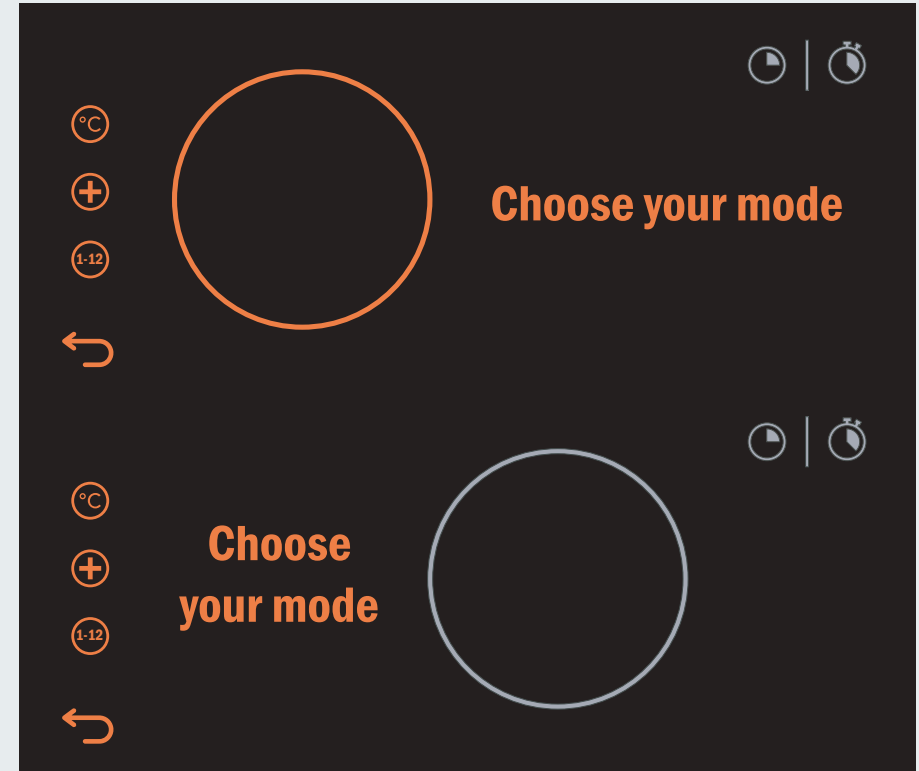


Figure 10: The different ways in presenting the iris slider.

Results

Although the opinions were a bit divided, most test persons thought the iris slider on the left was more logical, especially the choose your mode screen. Since the buttons and text are next to each other. But having the iris slider on the right looked better due to symmetry and less cramped buttons.

Most of the test persons thought the text ‘Choose your mode’ was not necessary and might be confusing. They said they only need it for the first use, but after getting used to the hob they do not need text anymore.

People did prefer the iris slide to be grey when not active, so it cannot be confused with a button.

Symbols and text

Are the symbols clear in what they mean and is the text neglectable?

Why

As mentioned in the feedback, some markets are not allowed to have any English in the design. Another point was that users may not need text if the symbols are known after first use. Because of the feedback, two small user tests were conducted. One about what the best symbols would be. The other is about the layout without text.

How

For the first test, 10 people were shown a paper with different symbols and asked which symbols they preferred and why. (Figure 11)

For the second test, 4 people were shown two printed-out layouts of the hob, one with text and one without. They were asked which layout they preferred and why. (Figure 12)

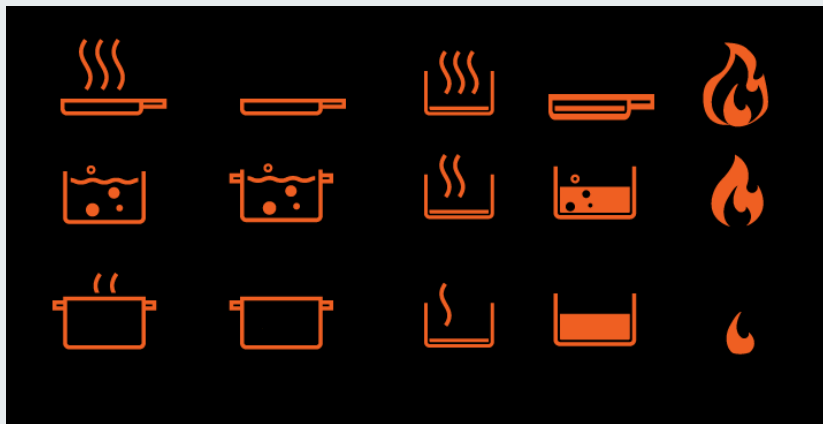


Figure 11: The variety of symbols.



Figure 12: Showing with or without text on the interface.

Results

For the symbols, people preferred to use the same pan, the pan with handles and steels to make it look more like a pan. People preferred three heatbars but then for lower temperatures smaller heatbars.

For the second test, people would like some text as an explanation for the first use, but after some time it is not necessary anymore. They said icons would be more easy to use because the text would guide too much. 'If I am not boiling but in the boiling zone, I do not want it to say I am.'

Timer function

Which timer function would you prefer? How can the timer function be used if used for long during slow cooking?

Why

As mentioned in the client feedback, they were doubting if the timer bar inside of the iris slider would be an improvement and if the bar would be as effective as short-timer use for long-timer use, as it will not show precisely. In other feedback moments, the text for the timer was not clear.

How

A test for the timer was conducted to get answers to the research questions above. This test was conducted on a computer using a lo-fi prototype on Figma. (See Figure 14 & 13) Nine people in the faculty of IDE were asked to look at two different layouts of the timer. They were asked which indication of the timer they preferred and why.



Figure 14: Prototype testing of the timer function.



Figure 13: The variety of visualising the timer on the interface.

Results

For the timer function, people preferred the bar as a countdown. They liked that the timer disappeared and you could click on it when needed. The letters are not understood and people said to just remove them.

For slow cooking, the preferences were diverse, since indeed the single bar visualisation is not as precise. People would like to have it seen in a segmented bar or would be more satisfied with the countdown.

Celsius mode

Can celsius be intergrated within the PLUS cooking? So the user only has two modes?

Why

As mentioned in the feedback, people were questioning why there would be a function for the PLUS cooking, only using temperatures, and the celsius cooking, also using the probe. They mentioned these two modes could be combined into one mode, to make the hob more easy to use. To see if this would indeed improve the hob a small test was conducted.

How

To test if the two modes combined would be better then separate, two lay outs were printed out. (See Figure 15) Three people within the IDE faculty were asked to look at the layout and see which they preferred. Most importantly, why?

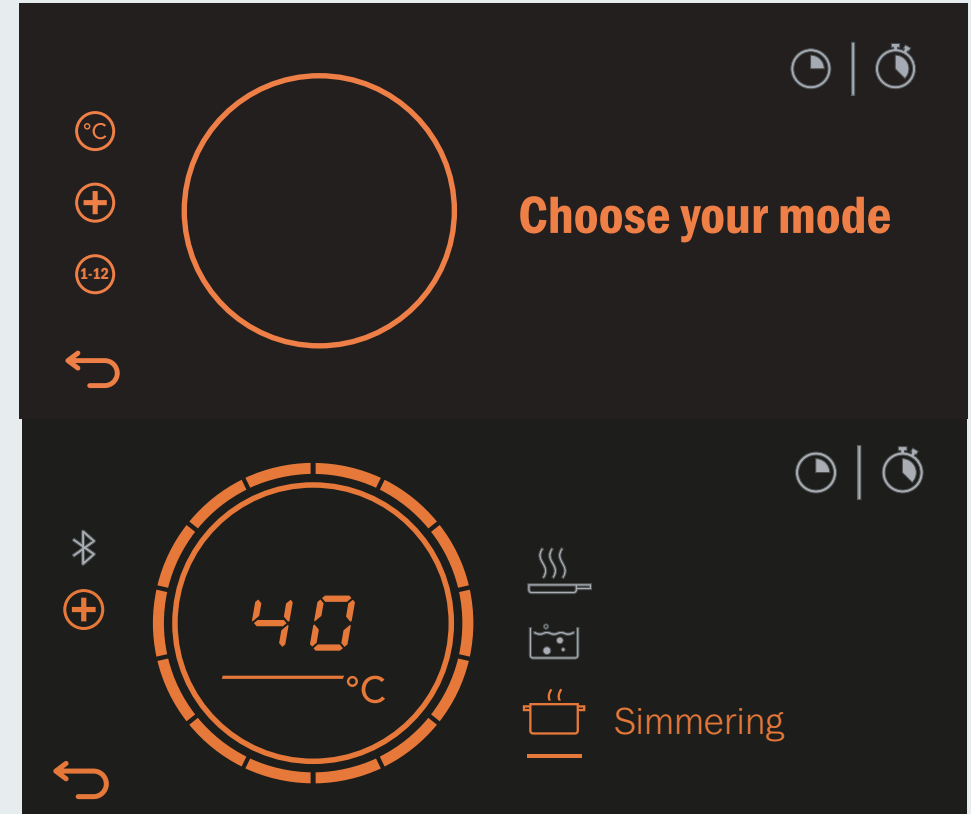


Figure 15: The integration of the celsius mode with a bluetooth icon.

Results

People were confused by the PLUS icon, only after the explanation they could understand what the mode was for. They suggested using the Celsius icon for the two modes together.

People were confused about the two modes and did not understand the difference. After hearing it is only the Bluetooth connection, they would rather see the modes combined. Thus, the Bluetooth icon is more intuitive than the celsius mode.

The Bluetooth icon felt like a conscious choice to make and signifies that a connection can be made, this is very intuitive.

Heat indication

How is it best communicated to show how 'hot' the celsius is? Is this valuable to connect with the power levels?

Why

A feedback point was that people would be a bit confused about the indication of the temperature. Is it the hob's temperature? And how do you know how 'hot' celsius actually is? People suggested showing in the PLUS mode an indication of how 'hot' the temperature actually is. To solve this feedback the idea arose to integrate the power level visualisation in the Plus mode. This is tested at the IDE faculty.

How

To test if the new idea was better than the previous version. Both layouts were printed out and tested on six people within the IDE faculty. They were asked which layout they thought was most logical and intuitive and why.



Figure 16: Above showing without heat indication, below image depicting with heat indication.

Results

People overall liked the indication of how 'hot' the temperature is, but some did comment the circle was not fully filled and looked like it was not ready to use and commented that maybe after a few times of using the hob, they learned how to use the temperatures. They also mentioned the icons were helping with the indication.

Temperature increment

What is the best temperature increments setting?

Why

As Björn once mentioned in Phase 2, increments of 5 in the Plus mode might be hard to use. Going from 0 to 160 are a lot of steps. Therefore a small test was conducted to find out if people would better like increments of 5 or 10.

How

A small paper layout of the Plus mode was shown to 10 people. They were asked whether they liked increments of 5 or 10 better and why.

Results

People thought for some occasions increments of 5 would be very specific. 'I do not know if I want to cook my pasta at 100 degrees or 105'. However, if you need more specific temperatures increments of 5 would be nice. They concluded increments of 5 would keep consistency on the hob and prevent frustration of the lack of control in Plus mode if you learned how to cook with celsius.

Conclusions & Design Refinement

To conclude from all the tests done, the following lists needs to be refined in the redesign.

- The placement of the iris slider will stay at the left, the defaults on the right.
- The symbols will stand alone and all text except 'boost' will be removed and there will be a card explaining the icon for first use.
- The icons will be changed to the icons that were mostly picked as intuitive in the test.
- The letters indicating hours/minutes/seconds at the timer will be removed.
- The visual count down will be used in segments for timer of 2 hours or longer.
- The Plus mode and celsius mode will be combined in to one, called temperature mode, using celsius icon as it is more intuitive.
- The probe can be connected with pressing a bluetooth icon.
- With using temperatures there will be an indication, just as the power levels, how 'hot' a sudden temperature is.
- The increments of the temperature will stay at 5.

In Figure 17 and 18 the refinements are shown. This is the redesign tested in the final redesign test.

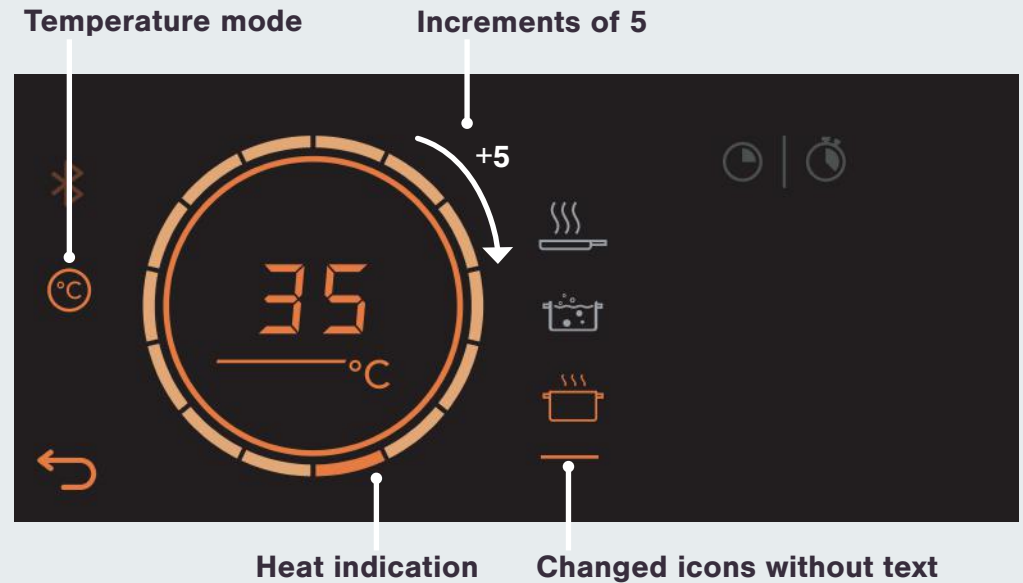


Figure 17: Design refinements

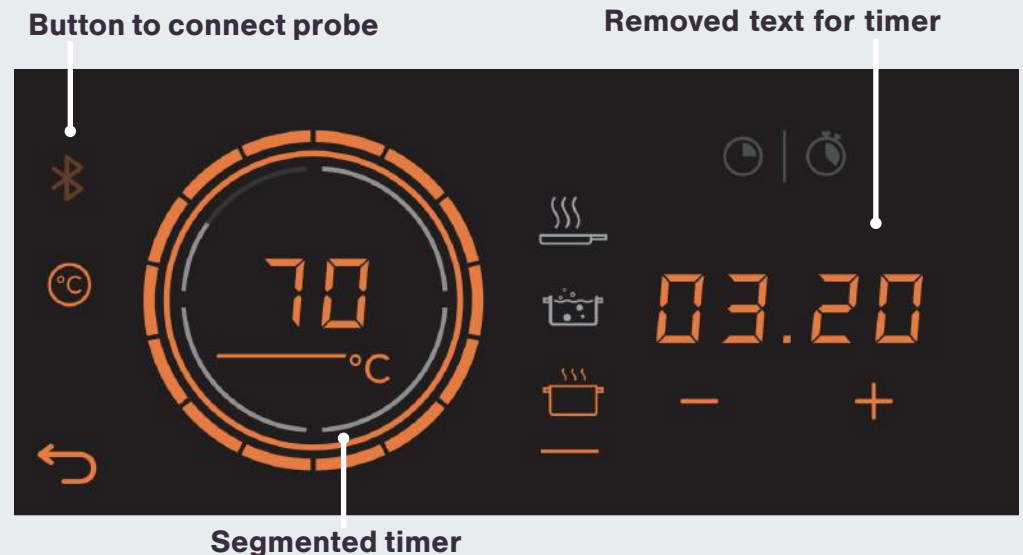


Figure 18: Design refinements Timer

3.

Design Evaluation

Once the design has undergone final refinement, the final redesign is evaluated to assess how potential users might perceive and experience it. By using various evaluation methods and conducting additional iterations, we aim to validate our design decisions, ensure that our design goals are met, and that the user experience has improved over the original.

Quick Chapter glance:

Cognitive Walkthrough

Pilot Test

Pre - User test

Test approach

Final User Test

User Test Analysis

Evaluation of testable targets

Conclusion



Cognitive Walkthrough

What

The cognitive walkthrough is a method for evaluating usability. A team member with the least bias, who was not involved in developing the prototype, is tasked with thinking from the user's perspective. Throughout the walkthrough, the 'user's' anticipatory and responsive behavior and actions are observed. See the conducted cognitive walkthrough in Figure 20.

Why

This method helps to streamline the design and prepare it for testing. It identifies any remaining usability issues so that the design prototype doesn't have obvious errors during testing. This allows for smooth testing and validation of design decisions. The process helps to pinpoint usability and user experience issues.

Outcomes

The outcomes were noted in a detailed table where we divided the outcomes into the anticipatory (feedforward) and responsive (feedback) behaviours and if they bumped into difficulties or errors why they did so (see Figure 19 for the related questions to determine the behaviours). A detailed description of the tasks with corresponding feedforward/feedback and difficulties is presented in Appendix A. The new usability (with improvements included) is compared to those that arose in Phase 1 related to our scope of the temperature cooking shown in Figures 21 & 22.



Figure 20: Execution of the cognitive walkthrough.

Feedforward: Will the user know what to do at this step?

Feedback: If the user does the right thing will they (1) know that they did it right and (2) made progress?

Figure 19: Questions to determine anticipatory and responsive behaviour.

The fixed set of temperatures of the presets in PLUS felt limiting

Preset icons were not clear (no connection between Celsius and PLUS cooking).

Letters on hob not clear

The amount of functions was overwhelming

Celsius cooking icon not clear & No clear connection cue for the Celsius Cooking accessories.

Could not easily switch between cooking modes

The countdown of the preheat stressed the user

After an error appeared, one had to restart to whole hob

The manual is not easy to read to find specific topics

There was no clear pre-heat indication, the PLUS cooking locked you within the pre-heat

Figure 21: The original interface.

Phase 1

A clear connection cue for Celsius cooking, *The blinking of the bluetooth should be slower*

Intuitive timer indication

Clearer icons

Letters only show after active pressing the timer

Less function with one temperature mode

A back button always available (except during pre-heating) for easy mode switching and error handling

Clear pre-heat indication that shows it is in boost.
The waiting time before going into pre-heating needs to be increased.

Freedom in setting the temperature.

A quick glance guide for easy understanding without manual, and easy access during cooking.

Figure 22: The redesign interface.

Phase 2

Pilot test

What

Before diving into the user test, we tested the test. We did this with one of our coaches, so a expert designer. The test was in the real test context (compared to the cognitive walkthrough). We approach the session as we had imagined to approach the user test. See a photo of the pilot test in Figure 23.

Why

The pilot test allowed us to make sure our test setup and prototype worked well. This allowed us to see if the taskflow was logical, that we got the insights we wanted from the test, to eventually get the most out of the user test as possible and do some last refinements before starting the test.

Outcomes

The completed observer form, which is the form template we set up for doing the user test and write down the notes (see for more explanation on set up for the user test next chapter) can be seen in Appendix B. Doing this pilot test led to the following insights and takeaways:

- Turn off phone notifications
- **To:** Let the user interact with interface with no distriction and as if it was on the hob.
- Have pre-heat for all temperature.
- **To:** Let the prototype work seamlessly, even if they decide to pre-heat on another temperature than asked in the test scenario.
- Have a more distinct change in task scenario's
- **To:** Allow for more in depth analysis of the different functions.
- Remove truning on the hob as a seperate task, but integrate it in the choose temperture mode task.
- **To:** Smoothen the task flow, since turning on the hob is not integrated in our prototype, so was not fluent integrated in the task flow
- Look into allowing user to use multiple hobs.
- **To:** Allow for userflow as if the user was cooking at home, so more intutive cooking. Limmitation would be that we will need more phones.
- Start another task while waiting for eggs to finsish cooking, start next task and then go back as timer is about to be done.
- **To:** Have a good flow in the user test, and have the user react when the timer was completed
- Write number indications at the PrEmo scale.
- **To:** Make it better to make notes for the observer, and easier for the participant to appoint.



Figure 23: Pictures of doing the pilot test.

Pre-User test

What

Before the final user test, we conducted a pre-user test as part of our preparations. Everything was set up for the final test and this was the final run-through before the participants arrived. We ran this test exactly as we would for the real test, complete with scripts and official setups (see Figure 24&25).

Why

This test is done to evaluate the test and design modifications after the adjustments of the pilot test and the cognitive walkthrough. It was intended to simulate as much as possible of the final user set-up, in order to detect the last fixable usability issues.

Outcomes

The main takeaways of the pre-user test:

- Have the **minimum amount of team members** in the test space, as we could give hints and influence the users with our presence.
- Reassure that **our redesign is tested** and not the participants, and with that, they should interact with the hob **as if they were at home**.
- **Change the order** of tasks to make the test more coherent and fluid
- Solve some Figma issues to make the intended interaction as smooth as possible
- **Adjust the wait time** for the appearance and disappearance of certain icons/buttons as they were overlooked.
- Ask the user to **rate the PrEmo** after every task.



Figure 24: Detailed picture of the pre-user test.



Figure 25: A overview picture of the pre-usertest.

Test approach

Research Questions

The two main research questions regarding whether the interaction of the Atag induction hob has improved compared with the original one:

- **Have the original usability problems identified in Phase 1 with the original product improved?**
 - **Patterns:** What is the user flow of the participant using the redesign?
 - **Experience:** How did the participants perceive the overall experience with the redesign?
 - **Usability:** In what way are the original usability problems improved, and how are the improvements received by the participants?
- **To what extent do new usability problems show up?**
 - **Usability:** Which aspects of the user flow using the redesign are perceived as unsatisfying?

The remaining research questions test whether the **interaction qualities and our design goal** were met (see Figure 26 and 27):

- **Intuitive:** Is operating the redesigned Atag Cooking hob intuitive? Did it respond as expected, and was the interface easy to understand? Was the user flow smooth without much hesitation?
- **In control:** Is the redesigned Atag Cooking hob designed to provide users with a sense of control, making it easy to access functions, manage their time effectively, cook in their preferred style, and use multiple functions simultaneously?
- **Guiding:** Does the redesigned Atag cooking hob assist users by providing clear feedback on their actions, allowing them to rely on the hob to guide their cooking process effectively?
- **Exciting:** Is the redesigned Atag cooking hob designed to offer an engaging cooking experience, allowing users to prepare food to their liking and encouraging them to explore new functions and unleash their culinary creativity in the kitchen?



Figure 26: Design goal



Figure 27: Interaction vision and qualities

Design criteria and testable targets

During the final Phase, we will assess the success of the redesign process by evaluating testable targets that are based on design criteria and research questions. These targets will be categorized into user experience, interaction, and product. The assessment will determine if

the targets have been achieved and if the redesign process has been successful. It's important to note that the targets for the UEQ may not be exact numbers, but rather estimates, as the test conditions vary significantly, making direct comparisons difficult..

Design Criteria

User experience

1. Users feel guided in their cooking process.
2. Users should experience that they understand how the hob works, and feel in control during their workflow.
3. Users should feel confident about their decisions when interacting with the hob.

Testable targets

User experience

- The feedback of the hob should be informative and actively show the user what the hob is doing. (1)
- Users could operate the primary use (powerlevel cooking and temperature cooking) without the need to consult the manual apart from looking at the quick glance menu. (1,2)
- Users should be able to understand the features without consulting the manual (apart from the glance). (2)
- The redesign should yield an UEQ score higher compared to the score from Phase 1, regarding the 'Plus mode' for Pragmatic Quality now called temperature mode. (2)
- The Temperature mode of the redesign should not score lower regarding the 'Plus mode' for Hediotic Quality, compared to Phase 1. (1,2,3)
- User experience with the redesign should yield an overall positive PrEmo score when compared to the original Atag Celsius Cooking hob when simulating the cooking process. (2,3)

Design Criteria

User Interaction

4. The hob's button placements and functionalities should be intuitive to navigate.
5. The feedback of the hob should let the user feel confident about their interaction, and help them understand what is happening
6. The hob should add efficiency and ease to the cooking experience
7. The hob should empower the user to improve their cooking skills.
8. Users should easily be able to choose their preferred cooking mode, empowering them to feel fully in charge.

Product

9. Users should feel confident in their understanding of the hob's elements (interface & accessories)
10. The hob has to be consistent (design, interaction and feedback) throughout the whole hob interface.
11. Temperature cooking should be a functionality that provides users with knowledge about cooking with degrees while still allowing room for exploration.

Testable targets

User Interaction

- Users move to the next step in their cooking flow with ease, they know what to do after their finished task. (4, 5, 6, 7, 8)
- In Temperature mode, users can explore and switch between preset default temperatures and adjust the temperature as needed. (6, 7, 8)
- The user can quickly check the timer progression on the interface. (6)
- The user waits during the pre-heat Phase. (5)
- The user can cook what they want, that fits their cooking style. (7,8)

Product

- The buttons (icon and placement) are easy to understand. (9, 10)
- Users should be able to understand their mistakes with the hob after one time, so when using it again after the first time this wrong attempt. (9, 10)
- Users understand the learnability of the function of the modes linked to the temperature. (E.g. frying happens at 140 degrees) (12)

Final User Test

What

A final usertest has been performed with 5 users. The usertest let the user interact with the redesigned interface of the hob. The test scope has a focus on the Temperature cooking mode.

Why

The user test is conducted to see whether our design goal, interaction qualities and testable targets are met. For a final validation to assess if the redesign improves the original design.

In this section, you will find the user selection, the test set-up, the test scenario's for the participants, the prototype, test limitations and data collection method.

Participant selection

To have the most valuable test, we set up participant criteria to ensure we were testing the intended target group of Atag within the market position. This helped us avoid practical issues such as a lack of experience working with an induction hob or not having the ambition to cook.

The participants are selected based on the following criteria:

- **Regular cooks** (more than 3 times a week)
- **Experience** with an **induction hob**
- **No previous experience** with the **Atag Celsius Cooking hob**
- At least three **different workfields**.

Criteria 1,2 & 3 are established to ensure target group fit, and criteria 4 is established to have different types of users that think differently regarding the design of interfaces or designs. The participant's profiles were as presented in Figure 28.



Experience level: 8 out of 10
Work field: Industrial Design
How often? 2-4 times a week
Cooking experience: 4-6 years
Familiarity with induction: Has one at home



Experience level: 7 out of 10
Work field: Industrial Design
How often? 2-4 times a week
Cooking experience: 7-10 years
Familiarity with induction: Has used it a couple of times



Experience level: 5 out of 10
Work field: Applied Mathematics & Education
How often? 2-4 times a week
Cooking experience: 7-10 years
Familiarity with induction: Has used it a couple of times



Experience level: 8 out of 10
Work field: Engineering and Policy Analysis
How often? 2-4 times a week
Cooking experience: 4-6 years
Familiarity with induction: Has used it a couple of times



Experience level: 4 out of 10
Work field: Industrial Design
How often? 2-4 times a week
Cooking experience: 11-15 years
Familiarity with induction: I cook with it elsewhere at least 1 time a week

Test set-up

What

We conducted the test at the hobe itself, where we displayed kitchen ambiance elements like spice racks. Our updated interface was shown to the user on a black phone placed on the stove. Participants cooked using paper print-out ingredients during the test. You can refer to Figure 29 for the test set-up.

Why

We placed the scenario at the hob, using a black phone to demonstrate the redesign most realistically. We printed out images of food, including “cooked/prepared” and “burnt” versions, to create the illusion of cooking and to communicate mistakes in their workflows (see Figure 30). These would be used by the moderator during the test.

How

Before the user test, we created a detailed test plan for consistency. The setup plan, moderator script, and observer form are in Appendix C, D & E. During the test, we had one moderator and one observer in the room to minimize external influence (as learned from the pre-usertest). After the test, we conducted a reflective group interview. The moderator’s role was to guide the user through the tasks, create a comfortable environment, and encourage them to verbalize their thoughts. The observer worked in the background, recording the user’s actions, thoughts, and challenges. We switched roles to maintain objectivity, see for our role division Appendix G.

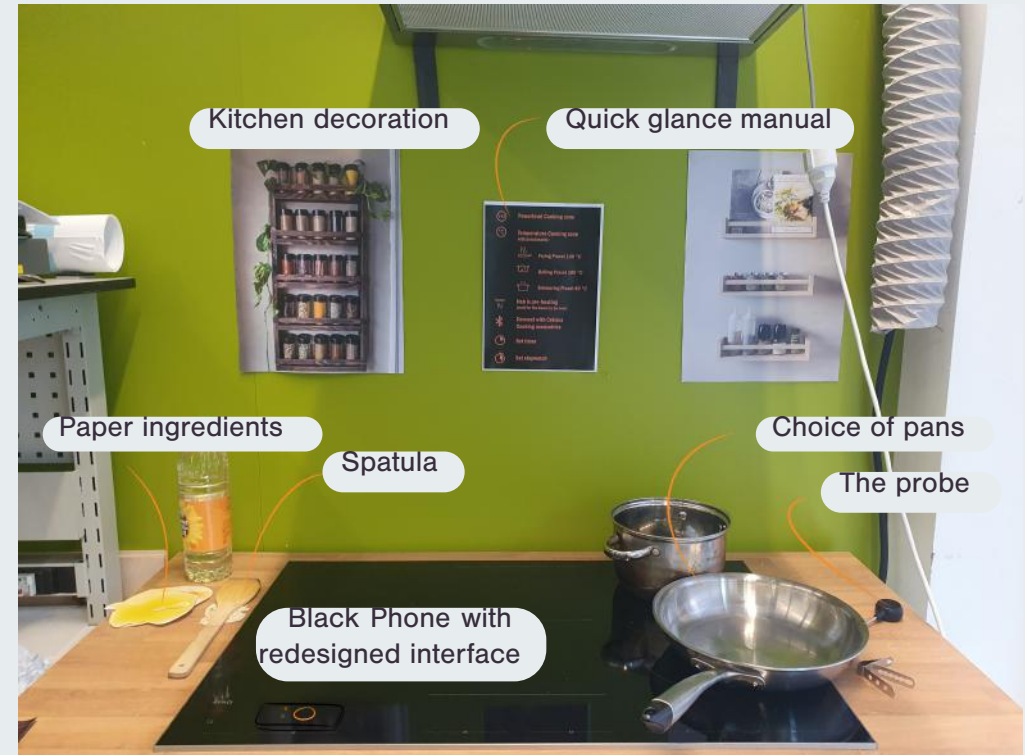


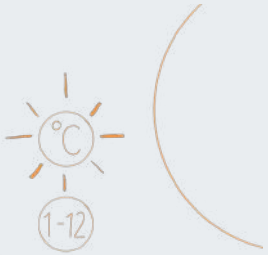
Figure 29: The user test set-up



Figure 30: Visualisation test scenario with paper cut-out ingredients.

Test scenarios

To evaluate our redesign overall experience, the users were given tasks that would guide them through our main improvements of the redesign. They were asked to do the tasks as described below. See Figures 31 t/m 33 for users in action during the first three scenarios.



1

Scenario: On a weekend day prone to explore cooking with temperature
Tasks: Set up the temperature mode
Objective: Examine the user's initial approach when asked to set up the temperature mode. Observe their interaction and determine if it is a positive experience.



4

Scenario: Boil an egg for 7 minutes.
Tasks: Boil the water and set a timer for 7 minutes.
Objective: Examine the user's initial approach when asked to boil an egg and gauge users' impressions of the timer functionality of the hob.



2

Scenario: Melt some butter for a butter sauce
Tasks: Set up temperature to 30 degrees and wait for preheat to add the butter.
Objective: To evaluate the ease of use and likability of the Temperature cooking including the benchmark points, and the understanding of the pre-heat function.



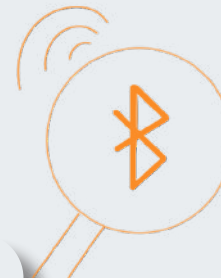
5

Scenario: Quickly roast some hazelnuts using the powerlevels.
Tasks: Set up the powerlevels and roast the hazelnuts.
Objective: Observe the ease of switching between cooking zones, and examine the users' reaction to the initial choice of cooking zones.



3

Scenario: Sauté onions and garlic.
Tasks: Set up the temperature mode to 140 degrees and add only oil before pre-heat.
Objective: To evaluate the use of the default temperatures and their learning curve regarding using the temperature cooking (including defaults and pre-heat). And capture their impressions.



6

Scenario: To explore the connection of the probe so you know how to set it up next time
Tasks: Connect the probe.
Objective: Observe the use flow of connection the Probe as integrated function in the Temperature cooking and determine if this additional function is well-received.

1



Set up temperature mode

Figure 31: Scenario setting up temperature mode.

2



Melt butter at 40 degrees

Figure 32: Scenario melting butter.

3



Sauté onions and garlic

Figure 33: Scenario sauté onions and garlic.

Usertest limitations

During our user test, we encountered **several limitations**:

- Due to our decision to limit the number of people present during the test, we were **limited to including the sound effects** of the hob, except for the timer sound.
- The **limitations of the prototype**, users were **unable to swipe along** the iris slider and they did **not have the visual cues** along the iris slider including at the end of the timer, the activation of the hob, and for the pan detection. This resulted in the initial pain points of the redesign, which were immediately resolved ones implemented in the original Atag cooking hob.
- Using **paper cut-outs** resembling raw and cooked food does **not directly replicate the experience of cooking on a real induction hob**. The use of paper cut-out ingredients may limit the overall enjoyment of cooking, affecting the Hedonic score related to feelings of pleasure. We assume that using paper for cooking inherently impacts the overall enjoyment of the experience.

Prototype

A digital prototype of the redesigned interface is developed in Figma to test the interaction flow of the redesign (Figure 34). The figma will be presented on a phone, as this is a familiar size to the size of the original Iris slider. When the user is using multiple pans on the hob, more phones will be put down alongside the hob, to mimic the actual interaction. Here is a link to a video containing how the various features are implemented. For a detailed explanation of the Figma flow for the realisation of interactive functions, see Appendix H.

The figma link to the prototype:

<https://www.figma.com/design/SCfNEt3hXEhXg4YrZm0boA/Cooking-Hob-Prototype?t=6sYFI2t41I4ITEAz-1>



Figure 34: Visual of test setting, front of video.

Prototype video link: <https://youtu.be/D0SezgMWHqQ>

Prototype Limitations

Function

Since the user interface is simulated on Figma, it cannot truly connect the hob's functions to the prototype. Therefore, users can only simulate cooking. Compared to our previous user tests with actual cooking, this might overlook many real-world emotional feedback factors, such as the anxiety a user might feel if the food burns during real operation.

Slider

Figma's functional limitations mean that each touch on the screen can only increase or decrease the temperature by 5 degrees. This does not fully allow for the Iris Slider's sliding selection to a specific temperature, which may result in a discontinuous experience for the user during testing.

Light

The prototype can only be tested on a mobile screen and not on the actual hob with indicator lights. There might be reflection issues during testing, making it difficult for users to observe changes in the interface.

Turn on/off

We focused only on the small, repetitive Iris Slider user interface rather than the entire hob user interface, so we did not create a power button for the hob. This requires manual simulation of the power on/off scenario during user testing, which may result in a discontinuous experience for the user.

Data collection

This study employed a mixed-method data collection and analysis, integrating both qualitative and quantitative approaches to investigate with triangulation the user experience of our redesign (see for a quick overview of elements Figure 35). By combining these methods, the study aimed to gain a holistic understanding of the user experience to see whether the original design has improved.

Quantitative elements

The quantitative evaluation methods that were used during the user test were UEQ and PrEmo (which both can partly be compared to the results of Phase 1), a Microsoft form with rates regarding the interaction qualities of the interaction with the hob and scores whether the presented task was easy - medium or hard to execute.

UEQ

The UEQ is a questionnaire established to measure the product experience of the user. This method was also used in Phase 1, so allows a general comparison of whether the design has improved on:

- **Pragmatic:** Indicates the usability of the product, consisting of user performance (e.g. effectiveness and efficiency) and user experience.
- **Hedonic:** Indicates to what extent users want to express themselves through the product and contributes to the personal development of the user.

For the comparison, we will look at the scores of Phase 1 concerning the PLUS cooking, as this was the core of our redesign. And focus on the improvement of the pragmatic interaction as our design scope focused on improving the user experience. See Figure 36 for the UEQ scales.

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UEQ, PrEmo, Surveys & Observer Data



Quantitative data

Observer Data & Behavioural actions



Qualitative data

Figure 35: Quick overview of the different data collection.

How did you feel about using the hob?

obstructive	o o o o o o o	supportive
complicated	o o o o o o o	easy
inefficient	o o o o o o o	efficient
clear	o o o o o o o	confusing
boring	o o o o o o o	exiting
not interesting	o o o o o o o	interesting
conventional	o o o o o o o	inventive
usual	o o o o o o o	leading edge

Figure 36: The ratings for the UEQ measurements.

PrEmo scale

The participants were asked to fill in a PrEmo scale after each part. PrEmo is designed to allow users to express their emotions by selecting one or more illustrations. The PrEmo scale allows the user to better define their emotion and also communicate this as all 14 emotions are represented by human emotions. See Figure 37 for the 14 emotions (during the test, the labels of emotions were removed so that they had to freedom to attach their own meaning to it. See Appendix I for a more elaborate explanation of the emotions.

Microsoft forms

After the user test, the users were asked to rate their experience from 1-6 with the redesign based on the research questions that arrived from our interaction vision with the qualities (intuitive, in control, guiding, and exciting). Please refer to Appendix F for the complete form and an example of the rates used for intuitive in Figure 38.

Observer quantative data

The observers rated the **effectiveness of the task** (i.e., if the user was able to complete the task) between **easy, medium and hard**. This was done through a combination of quantitative and qualitative methods. The quantitative measure was the amount of time taken per task and the number of steps taken for the user to complete the task. The qualitative measure is taken into account through observing any evident frustrations the user faced while interacting with the redesign (see Appendix E for the observer template). In order to assess the **experience** of the product as **positive, neutral or negative**, the observer noticed the emotions and remarks of the user during the use of the product.

Qualitative elements

The qualitative elements of the research mostly consisted of the observation of the participants during the user tests. The observer noted in the observer from (Appendix E), the train of thought of the user and how they interacted with the interface. This aimed to find the answers to the research questions regarding patterns, experience, and usability. After the user test, we had a reflective interview supported by the Microsoft forms and the PrEmo scale. The reflective interviews provided nuanced insights into participants' perspectives and experiences.



Figure 37: The PrEmo 14 emotions.

Intuitive

This section tests whether the interaction with the hob felt intuitive.

1. You feel like the hob responds to the interaction as expected? ☐



2. The interface is easy to understand ☐



3. Cooking with the hob is smooth and doesn't give you any hesitations ☐



Figure 38: Example of the rates to derive the answers of the research questions regarding the interaction qualities.

User Test Analysis (Qualitative)

Qualitative data was gathered by the **observers** who took notes during the user tests which show recorded emotions, **quotes by the participants**, changes in body language, time and steps taken to complete each task. These **notes** were then **compared and analyzed** to reveal some interesting and valuable insights, all of which can be seen in Appendix K. The most important ones are shown below, per task:

Task 1: Choose temperature mode

Strong point

- It is an intuitive and simple interface because of the quick glance menu.



"Doesn't feel like there were too many menus to go through."

Pain Point

- It was not clear the Iris slider was on as it remained grey instead of orange.



"If it looks like this I don't know if its on or off."



Task 3: Add oil and fry onions at 140°

Strong points

- It was easy for them to set the temperature by using the frying preset.
- They liked the freedom of changing the temperature in increments of 5 after jumping to the presets.
- The icons were clear due to the accompanying text and the quick glance menu that allowed them to learn quicker than a manual.

 "I remembered the instructions (about preheating and temperature) from the quick start guide."

Learning Curves

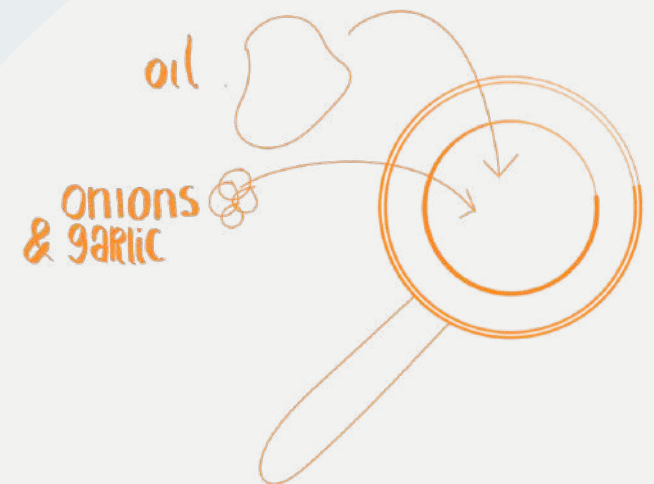
- At first use, 2 out of 5 did not know that they could change the temperature but afterwards learnt it for the next tasks.
- The quick glance did help them understand the presets/default settings, but cooking with it helped learn it further.

 "I didn't do anything special. It felt like a normal day of cooking."

Pain Point

- One found it to be a lot of tapping due to the limitation of the prototype where they couldn't slide/scroll instead.

 "Oh now I have to tap a lot, more than what I'm used to with an induction hob."



Task 4: Boil an egg at 100° and set the timer for 7 minutes

Strong points

- The timer sound was considered to be distinctive.
- They intuitively pressed the timer button to check the exact time, after it disappeared.
- 3 out of 5 participants liked the visual grey bar for estimation of time left.
- 2 out of 5 participants liked that there was no active countdown (numbers) on the hob all the time.



"The hob guides you, you don't have to do it all by your self."



"Ah this timer visual is nice. When I return from the toilet, it is a nice way to quickly look to see how the hob is doing."

Learning Curves

- 3 out of 5 users did not notice the visual grey bar was there initially and did not expect that the numbers would disappear.
- One user didn't know that the timer would automatically started if you waited (instead of pressing a start button for it to start).

Pain Points

- Visual cues are needed for the completion of the timer, to know which timer is finished for which cooking zone.
- One user felt it was unclear if the timer was still on when the numbers disappeared.
- One user pressed the 'H.M.' to change the Hours and Minutes to Minutes and Seconds.
- Non-visual users (2 out of 5) wanted to have the numbers to be always there.
 - This was also due to their preference of having definite numbers as opposed to a vague visual.



"If there was a setting to see the timer (numbers) always, I would prefer that."



Task 5: Quick roast hazelnuts with power levels

Strong points

- It was easy for all of them to use the power levels due to familiarity.
- The back icon was easy to understand for 4 out of 5 users as it was considered to be a universal symbol.

 *"If I was quick and in a hurry, I would just put it to a high power level like I normally do on my induction."*

Pain Points

- It wasn't intuitive to know how hot the power levels were (specially after using the very precise temperatures).
- One user didn't realize the back button and said he would rather want the buttons of the modes (temperature mode and power level mode) to always be there.

 *"It is nice if both modes are always there."*

Learning Curve

- They didn't understand which power level to use for roasting.



Task 6: Connect the probe



Strong point

- The blinking of the bluetooth icon was familiar and clear that something is being connected to the hob.



"It is intuitive because it looks similar to other Bluetooth connection things."

Pain Points

- It was not clear what the hob connected to - the probe or the pan.
- There was a lack of feedback of the connection being completed.



"How do I know if it is connected to the probe, if there are multiple things you can connect to?"



"I expected something to happen" (on the hob, when it connected with the probe)

Conclusion (Qualitative)

The quotes on the right (in orange), showcase the overall feeling of the participants' use with our improved prototype, which is mostly positive. While some parts did require a bit of learning, most tasks were done with ease, clarity and confidence. However, tasks 4 and 6 showed us some pain points that needed to be addressed.

For the final round of refinements, as seen in the upcoming chapters, our **focus** will be on developing the **timer function (Task 4)** and the **Bluetooth connection (Task 6)**.



"Using it once was intuitive. Second time will be easier. I will fly through."



"It is new but also interesting to see how it works."



"I like the temperature more, as you feel more in control and feel more like a scientist, like a pro chef."

Quantative Analysis

UEQ

After the user test each participant was asked to fill in a UEQ about the overall use of the temperature mode, to find out what their opinion about the overall use is. The participants were told to think out loud during the UEQ. The filled in UEQ's can be found in Appendix K. In Figure 39 the average outcome of the UEQ can be found.

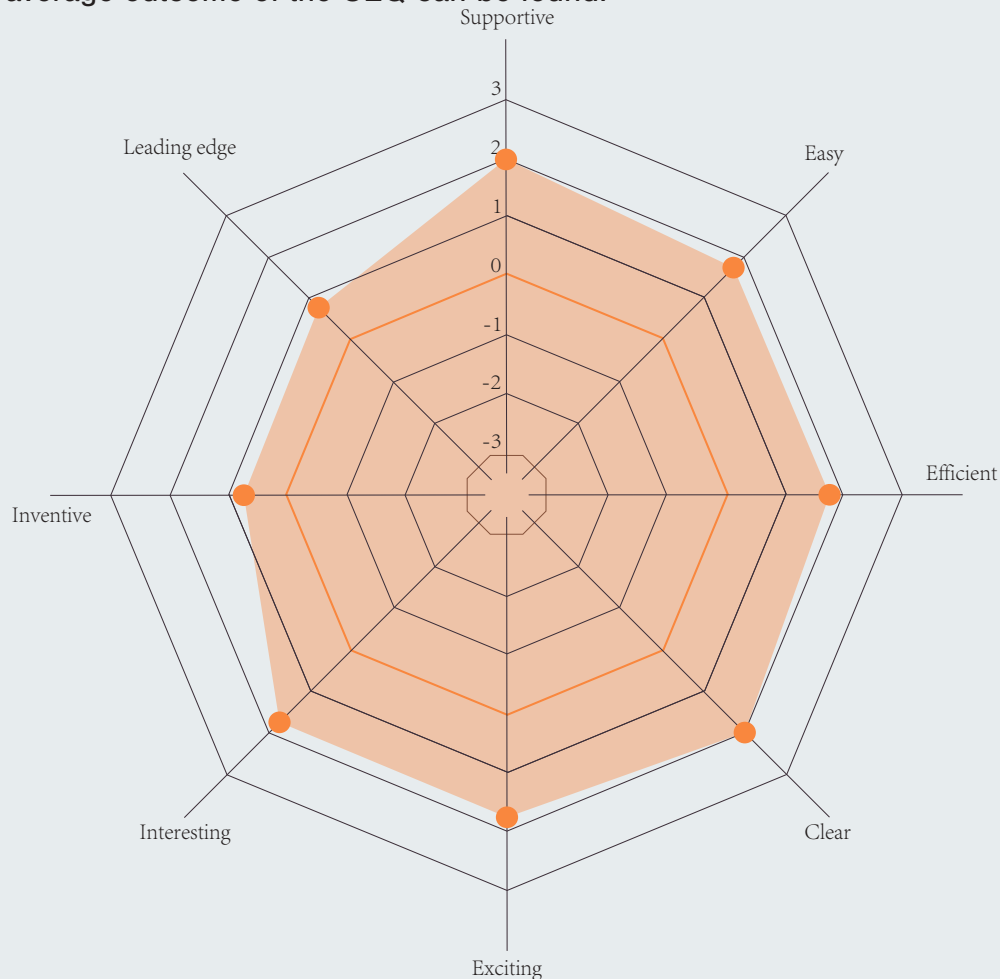


Figure 39: The depiction of the results of the UEQ.

Comparison to the initial design

The choice to let the user fill in the UEQ for the final redesign test was to make a comparison to the initial design. In Phase 1 a user test with the initial design was conducted. Participants filled in UEQs for each step of the user test. A UEQ for using the Plus mode was filled in, exactly the mode we tested in the redesign. (See Appendix L)

Although the UEQs of both tests are about the same modes, it is not one on one comparable. It is kept in mind that the test that was conducted is not exactly the same and can influence the answers of the UEQ. In the test conducted in Phase 1, participants needed to cook an actual meal, while in the test conducted in this Phase cooking is simulated with the help of paper ingredients. We think especially hedonic qualities (exciting, interesting, inventive, and leading edge) are influenced by these differences.

Conclusion

To conclude, if the redesign changed the interaction with the hob, only significantly large differences have been looked at. A significant difference is stated in a difference of mean than more than 1. Keeping this in mind, especially the pragmatcal qualities improved in the redesign. The redesign scored more than one point better on supportive, easy, and clear.

PreMo

After each task of the user test, the participants were asked to point to one or more emotions on the PreMo scale and explain why they feel that way. The used PreMo scale can be seen in Appendix I. In the graphs below the experienced emotions per task are visualized.

Task 1: Choose Temperature mode

For task 1, all participants felt positive emotions, see Figure 40. They said they are curious to see how the hob works and are eager to learn.

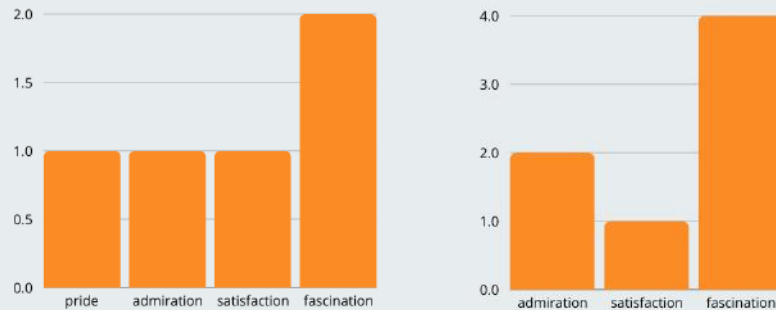


Figure 40: The PreMo results for Task 1 and 2.

Task 2: Melt butter at 30 degrees

Task 2 made the participants fascinated, as can be seen in Figure 40. This is because it is a new and innovative way of cooking. It is an exciting way of cooking and they are curious to see what more is possible.

Task 3: Fry onions at 140 degrees

With this task people felt mostly proud (see Figure 41), because with the gained knowledge in the last task they could do this task with confidence. They praise the product for being easy to learn.

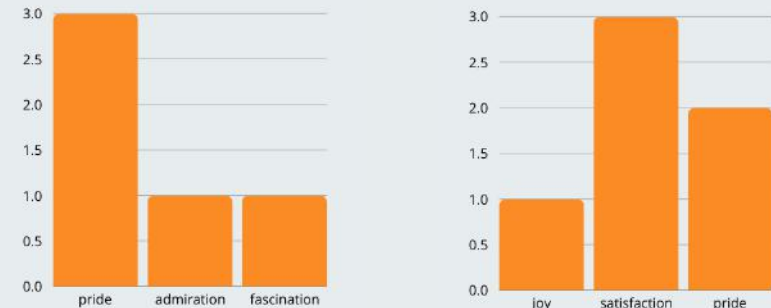


Figure 41: The PreMo results for Task 3 and 4

Task 4: Boil an egg and set a timer

During this task participants felt they were guided by the hob, the presets are an efficient way of getting to 100 degrees (Figure 41). Using the timer went smoothly as well and did not cause any confusions.

Task 5: Roast hazelnuts in powermode

The participants felt a mix of positive and negative emotions (Figure 42). This mode is boring compared to the temperature mode. In one case, the go back button in our prototype was not working, which made him feel diss

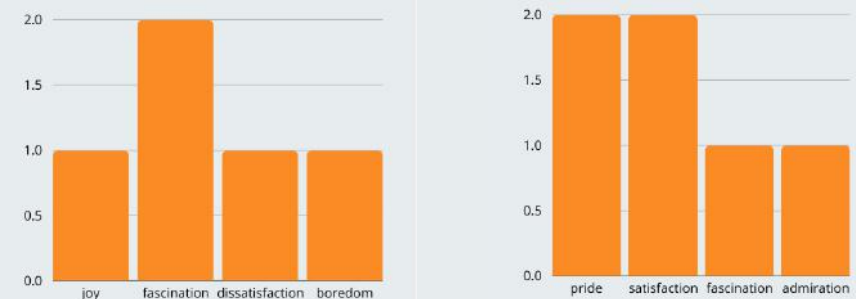


Figure 42: The PreMo results for Task 5 and 6.

Task 6: Connect probe with Bluetooth

Connecting the probe with bluetooth is easy and feels intuitive, according to our participants. They are proud to immediately understand what they need to do (Figure 42).

Google Forms

After each user test, the participants were asked to fill in a Google Forms questionnaire. In this questionnaire we ask the participants to rate our redesigned hob on a scale of one to six, about the previously defined interaction qualities. The complete questionnaire can be seen in Appendix F. These qualities describe the intended interaction with the hob, the results from this questionnaire are evaluated here:

Intuitive

For this section of the questionnaire we asked the participants whether they felt the interaction with the hob was intuitive. The following questions were asked:

1. You feel like the hob responds to the interaction as expected?
2. The interface is easy to understand?
3. Cooking with the hob is smooth and doesn't give you any hesitations?

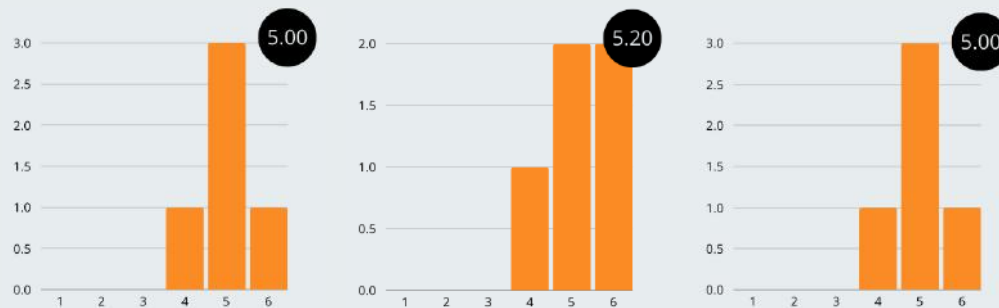


Figure 43: The intuitive scores for question 1,2 & 3 respectively.

All 3 questions were rated with an average of 5 or higher see Figure 43. This means all candidates would say the redesign is intuitive to use.

In control

With this section we want to determine if using the hob makes you feel in control. This was done by asking the following questions:

1. You could access functions easily?
2. You felt in control with your tempo of cooking with the hob
3. The hob allowed you to cook in your way (your usual cooking style)
4. The hob allowed you to switch easily between multiple cooking modes (temperature and power level)
5. The hob allowed you to switch easily between multiple cooking presets (within temperature mode)

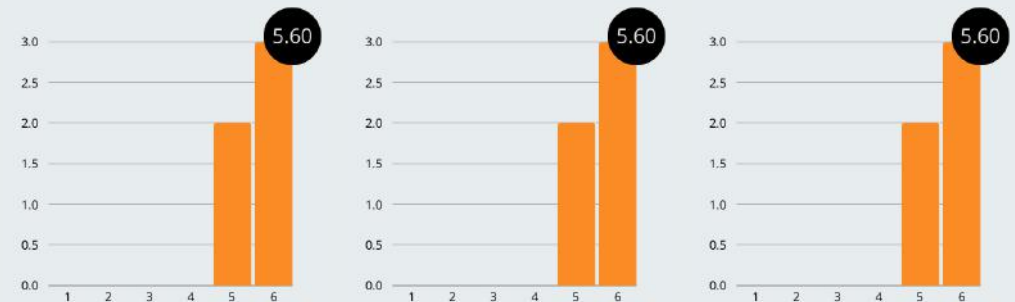


Figure 44: The scores for feeling in control for question 1,2 & 3 respectively.



Figure 45: The scores for feeling in control for question 4 & 5 respectively.

The participants rated the hob with an average of 5 or higher per question, with the lowest score on switching between the modes (Figure 44 & 45). The other questions are rated with an average of 5.60 and this tells us that the hob makes the participants feel in control.

Exciting

This section aims to find out if the participants think the interaction with the hob is exciting. This is determined through these questions:

1. With this hob, you can cook your food the way you like it
2. You see yourself experimenting your cooking style with the hob
3. The hob supports you in learning about new functions
4. You are eager to use the hob for cooking



Figure 47: The guiding scores for question 1 & 2 respectively.

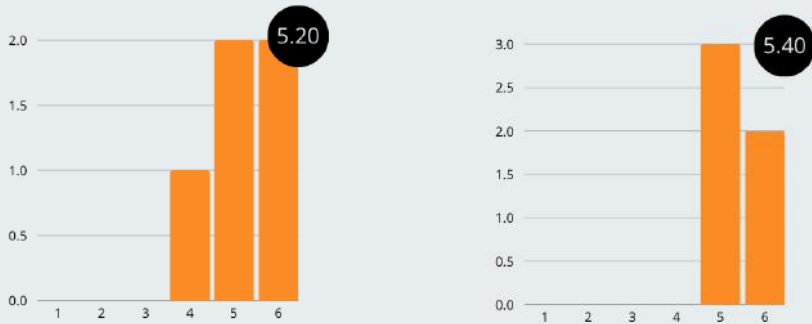


Figure 48: The guiding scores for question 3 & 4 respectively.

All questions in this sections have an average of 5.2 or higher, with one question even being rated with an average of 5.8 (Figure 47 & 48). The participants feel the product invites you to experiment with your cooking style and enables you to cook in your own way. This shows that interacting with the hob is exciting for participants.

Guiding

This section tests whether the hob guided you in the cooking process. The following questions test this:

1. The hob provided clear feedback on your actions
2. You could rely on the hob to perform as you instruct it to
3. The hob steered you in the right directions

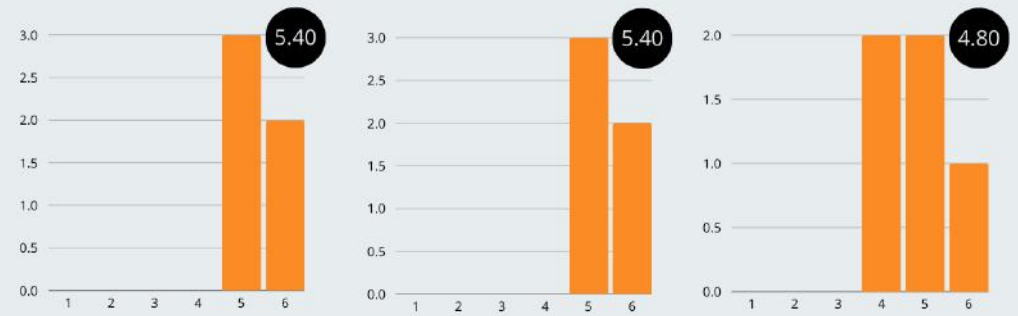


Figure 46: The guiding scores for question 1,2 & 3 respectively.

In this section, the lowest average is 4.80 (Figure 46). The participants felt that the hob did not steer you that much in the right directions, but this average is still above 3, and therefore sufficient. This shows users think the hob is guiding, but there is some room for improvement.

Conclusion

After evaluating the interaction qualities, it can be concluded that all interaction qualities are evident in the new design. The lowest overall score is 4.8 on a scale of 1-6, with an average positive score of 5.35.

Overall user test conclusion

The user tests have given us a great amount of valuable insights. We will take these insights to make some changes to the redesign. From the user tests we can see that overall, users feel positive during use. Mostly positive emotions were expressed, only within the power level cooking, participants felt that it was boring compared to the temperature mode.

The participants feel that the product is in line with our interaction qualities, as we can see from the results of the Google Forms. Some improvements can be made in making the hob more guiding than it currently is.

If we look at the UEQ that was filled in after each test, we can see that the redesign scores above 0 for every aspect. If we compare the UEQ from this user test, with the user test in part 1 we can see that the redesign is more supportive, easy, and clear.

From the user tests, strong points, pain points, and learning curves were derived. This showed us that some tasks require a bit of learning, but most tasks were done with ease, clarity, and confidence. The painpoints we found will be taken into consideration for the redesign, this includes the timer function and Bluetooth connection.

We take all these results with us for an iteration on the redesign, the refinements will include the timer function, Bluetooth connection, and making the hob more guiding.

Evaluation of testable targets

The testable targets that derived from our design criteria, as mentioned above in Chapter 3, were evaluated through the results of the final user test. Table 1 outlines the user actions that confirmed how and if the testable

targets (now presented in a summarised way, for a full description, please refer to Chapter 3) have been achieved. Eventually, make the first step on concluding on whether we reached our design goal with our intended interaction qualities.

Reached?	Testable Target	User Test Outcome
Yes	Users could operate the primary use without the manual	Users did not feel the need to ask for a manual.
Yes	The redesign should yield an UEQ score higher Pragmatic Quality compared to Phase 1.	The Pragmatic quality of the UEQ scores higher.
No	The Temperature mode of the redesign should not yield score a lower regarding the 'Plus mode' for Hediotic Quality compared to Phase 1.	The Hediotic Quality scored slightly lower.
Yes	User experience with the redesign should yield an overall positive PrEmo score	Users expressed only positive emotions throuhout all parts of the user test.
Yes	The feedback of the hob should be informative and actively show the user what the hob is doing.	Users knew what the icons and indications on the hob meant without questioning, or in need of the manual.
Yes	Users move to the next step in their cooking flow with ease	Users knew what to do at each step to move forward without questioning.
Yes	In Temperature mode, users can explore and switch between preset default temperatures and adjust the temperature as needed.	Users were able to switch and adjust, and expressed this was with ease.
Yes	User could quickly check the timer progression on the interface.	During the task of checking the time, the visual users noticed the grey bar and was perceived as nice, the less visual participants lacked the more precise information.
Yes	The user waited during the pre-heat Phase.	Not more than the expected learning curve of one mistake when preparing the meals, so they did wait.
Yes	The user can cook what they want, that fits their cooking style.	Users suggested they could cook as they normally would without limitation.
Yes	The buttons (icon and placement) are easy to understand.	Users knew what to do at each step to move forward without questioning and knew where to find the buttons without mentioning their hand was in the way or the placement was off
Yes	A mistake in interaction should only happens once. Users should be able to understand the interaction based on their mistakes.	Users did not make additonal mistakes after the sometimes accurance of a mistake (...times)

Table 1: Evaluation of the testable targets

As can be concluded from Table 1, only one testable target has not been satisfied. This is the testable target for the Hedonic Quality. This could be explained due to the use of paper cut-outs during the user test. Although, this approach provided valuable insights into the user experience flow. It's important to note that it may **not directly replicate the experience of cooking on a real induction hob**. The use of paper cut-out ingredients may limit the overall enjoyment of cooking, affecting the Hedonic score related to feelings of pleasure. We assume that using paper for cooking inherently impacts the overall enjoyment of the experience.

Conclusion on Research Questions

To conclude the set research questions for the user test. The two main research questions:

- **Have the original usability problems identified in Phase one with the original product improved?**

Yes, since the testable targets were met, the user flow did not give a lot of errors, and the overall UEQ for pragmatic quality has improved. It can be stated that the usability problems have improved.

- **To what extent do new usability problems show up?**

However, new aspects to improve upon the redesign emerged from the user test. These aspects are discussed in Chapter 4, along with other recommendations for the design and further testing.

We can conclude that the testable targets developed to reflect the design goal and criteria were achieved, thus **meeting the design goal**. However, it's important to consider the limitation of using paper cut-outs, which affected the Hedonic quality during the user test. All interaction qualities were recognized during the interaction, evident in the user flow, and supported by the outcome of the Google forms as outlined in the quantitative analysis.

4 ■ Final Design Detailing

The user test has shown improvement areas in our redesign. In this chapter, we sharpen our initial redesign based on the feedback we have gotten through these user tests. We present the personas for whom we will be redefining the concept and will present our final prototype proposal for both the hob interface and for a small part of the celsius accessories (the probe and the celsius pan).

Quick Chapter glance:

Final persona

Client meeting 2

Current interaction

Final Design

Final redesign interaction

Redesign sccessoiries



Final Personas

During the user testing, it came to light that the personas that were used during the first two Phases were not a complete representation of the actual users. We noticed that some information from the personas is missing or not giving a broad enough representation.

Atag's marketing personas

Before understanding what needs to be changed in the personas it is important to look back at the previous personas. In the first two Phases there were three groups from Atag:

- **The Social chef** : someone who likes to cook for an audience
- **The Semi-Pro**: someone who cooks to improve their own skills
- **The Pleasure Seeker**: someone who like trying out new things, but not in complete detail

These three groups we seen more as marketing groups instead of the actual users, as in real life the hob would not only be used by the purchaser, but also by their kids, guests, cooks, etc. It is **important not to confuse** the marketing personas with the product personas.

Our personas

Our redesign personas from Phase 2 are the **simple routine chef** and **the exploratory chef**. Although these personas are still used as a base for our redesign, an addition can be made to the personas. As it was noticeable during the testing some people preferred visuals, such as icons, and others preferred exact time, text, and degrees. Seeing actual time and text would also be preferred by new or irregular users who need a bit more guidance. Through this new addition, the following personas were created.

The visual exploratory chef

Michelle is an expired professional chef. Finally not having to cook under pressure, she loves to explore different recipes and different settings on the hob. She loves to be in control of the temperature and not guided and distracted by the hob too much with unnecessary text.



The non-visual exploratory chef

Jack just started with working as a teacher mathematics. In weekends he loves to cook for his family. He slow cooks often, always using the temperature mode to cook on the degree precicely. As numbers is what he knows best, he loves to see precise timers for slow cooking.



The visual simple chef

Anna is a single mom of three kids, after work she likes to spend as much time with her kids. Cooking is not here main priority, only when her kids are away. After work she loves to put together a simple meal with the power levels, without spending to much time on it, maybe her kids can help her cooking when they are a bit older



The non-visual simple chef

Tim works as a student hobby chef to cook for families who do not have enough time to cook. Tim uses the Atag Celsius hob once a week. As he does not have too much time to explore the hob, he always uses power levels to make simple dishes for the family. He loves to be guided by the hob



Client meeting II

Midway through the process, we had a midterm client meeting with four members of Atag. They were curious about what we have been doing so far and were curious to see what we got out of the user test so far. For us, this was also a great opportunity to reflect and put into words what

we have been learning from the user test so far. The meeting mostly consisted of a recap of Phases one, two, and three. However we got some insights while showing the previous reports, during the demo, and while discussing the user test results as shown below.

Demo

After showing our demo, they mentioned a few things:

- They really liked the switch between modes that changes the interface.
- The reaction to the freedom of choosing within the default temperatures was perceived as nice.
- They were wondering if we took the things separately.

Learning to cook

Atag liked our design goal of giving regular cooks the confidence to cook but they're concerned that our current default temperature settings may not provide enough knowledge about actual cooking. They suggested exploring the possibility of providing users with more temperature knowledge while maintaining the current intuitive interaction.

Implications

Make a quick glance menu on how to cook with temperature. Recommend for further testing to combine several items of the original and the redesign to see the biggest improvement impacts.

Current interaction

To show the improved interaction, first a storyboard of the current interaction is shown. (Figure 49) Later in the report the storyboard of the final redesign is shown. The phrases in the speech bubbles are quotes of the test participants in Phase one.

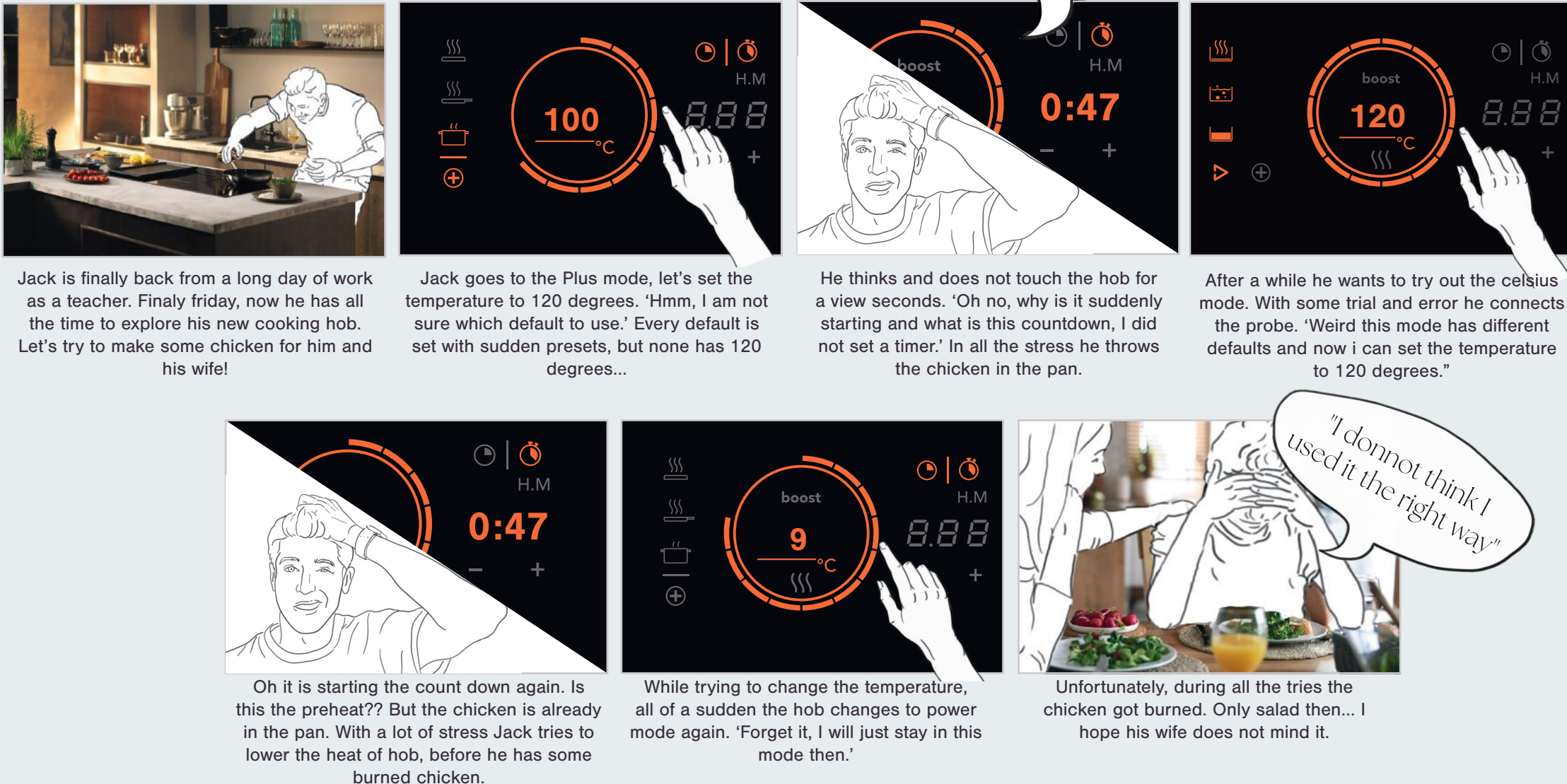
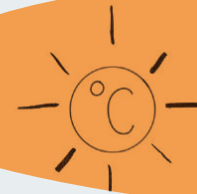
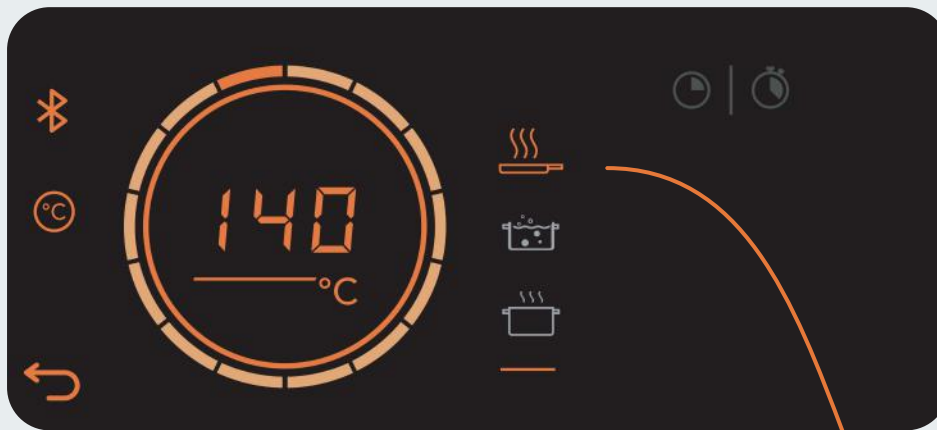


Figure 49: The storyboard for the original Atag Cooking hob. Please note that the hands demonstrating the interface interaction are not to scale.



Temperature Mode

Presets

Within the 'Temperature' mode, we have presets that help guide the user to understand which temperature can be used for which functions. There are three main presets as shown below and in Figure 62:

Frying (140° to 240°)

When the user clicks on the frying pan icon, the hob sets it at 140°, from which the user can increase or decrease the temperature in increments of 5.

Boiling (100° to 135°)

When the user clicks on the boiling pot icon, the hob sets it at 100°, from which the user can increase or decrease the temperature in increments of 5.

Simmering (5° to 95°)

When the user clicks on the steaming pot icon, the hob sets it at 40°, from which the user can increase or decrease the temperature in increments of 5.

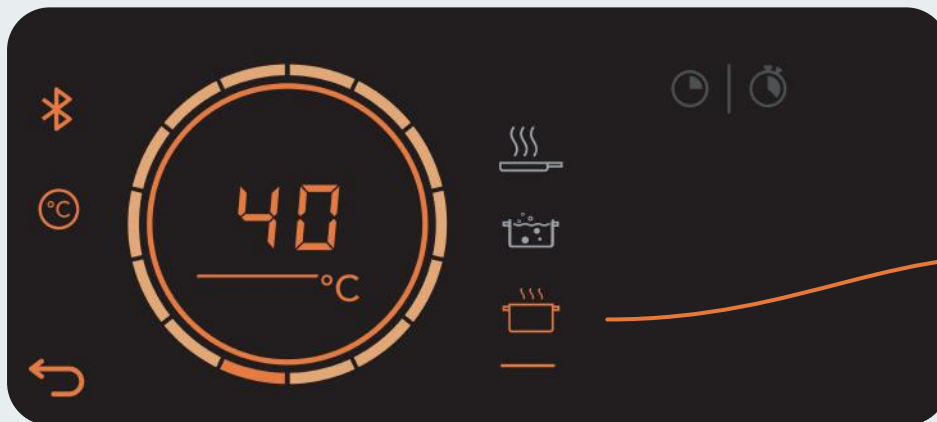
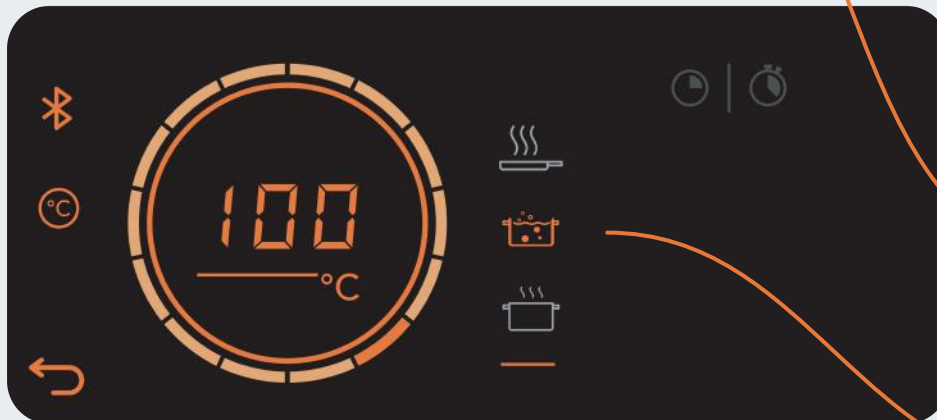


Figure 62 : Hob Interface - Temperature mode with Presets

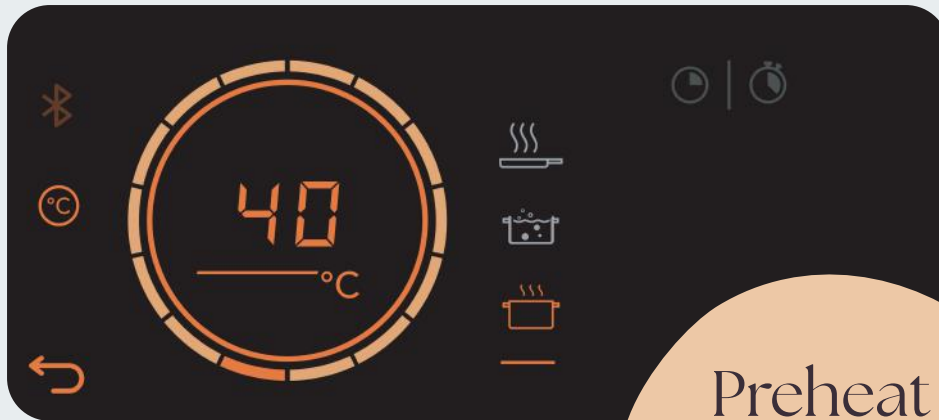


Figure 63 : Hob Interface - Right before preheating starts (Setting the temperature)

The preheating function has been redesigned to provide the user with visual information with regard to *what is happening* (the boost) and for *how long* (the progress bar). (Figure 64)

Boost

Here we show that the hob is using 'boost' mode to bring the temperature up to the required degree quickly.

Progress bar

While the hob does so, the user can see the amount of time left for it to be ready through the visual progress bar.

No Back Button

In order to ensure a smooth preheating flow, we have removed the back button here to enable the user to stay on this function instead of allowing them to change modes. Changing modes would disrupt the technological flow of the preheating, thus causing error.



Figure 64 : Hob Interface - Preheating function in Temperature mode

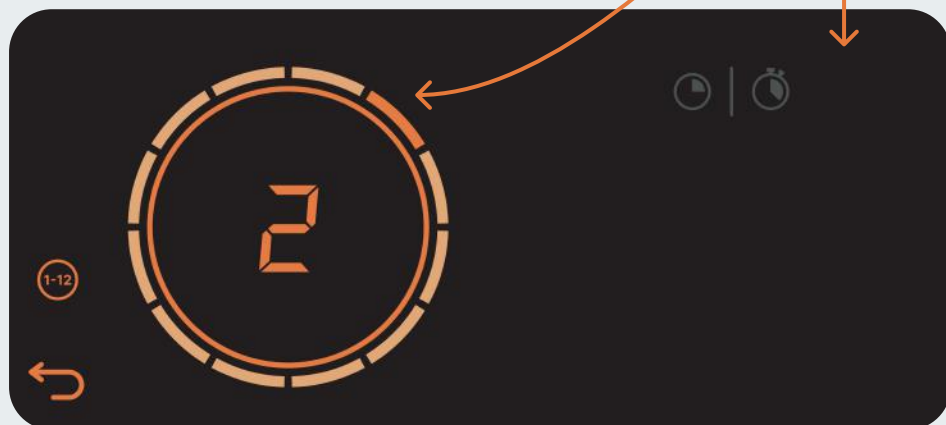
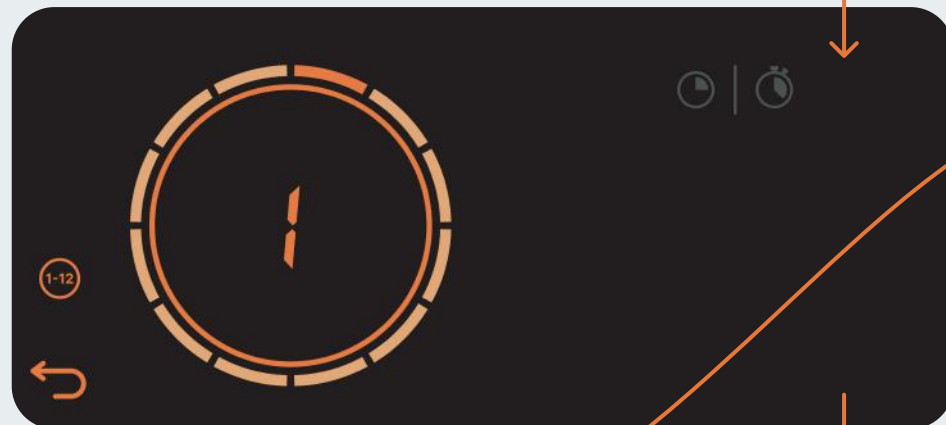
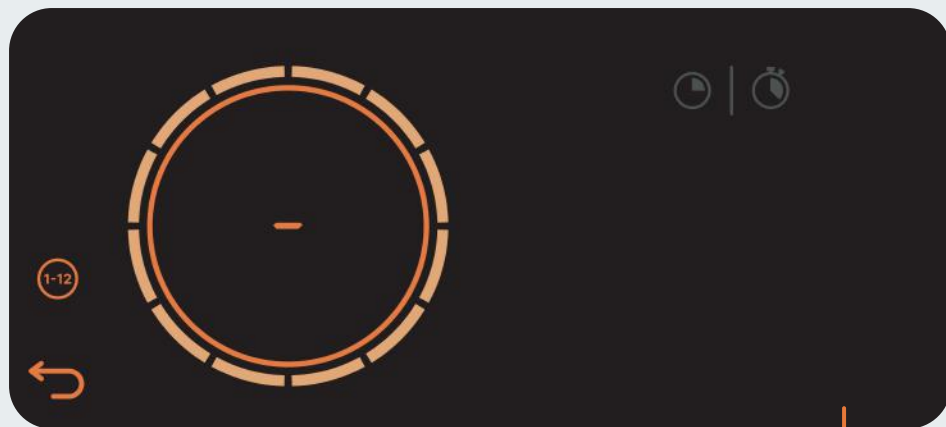


Figure 65 : Hob Interface - Power Level Mode - Setting a Level

Power Level Mode

The power level mode is now placed on the hob as a separate mode that can be accessed in the start, right after turning the hob on (Figure 61). Here is how it is used:

- Setting a level is the same as setting a temperature, where you slide along the Iris slider to chose a level.
- You can see the amount/level of heat visually with the individual orange segments on the slider. *(Similar to design of Iris slider for Temperature Mode)*
- Once the level is selected, after a second, the whole bar becomes orange to show that it is now set. *(Similar to design of Iris slider for Temperature Mode)*
- There is no preheating in this mode as the hob slowly brings it up to the chosen heat level.

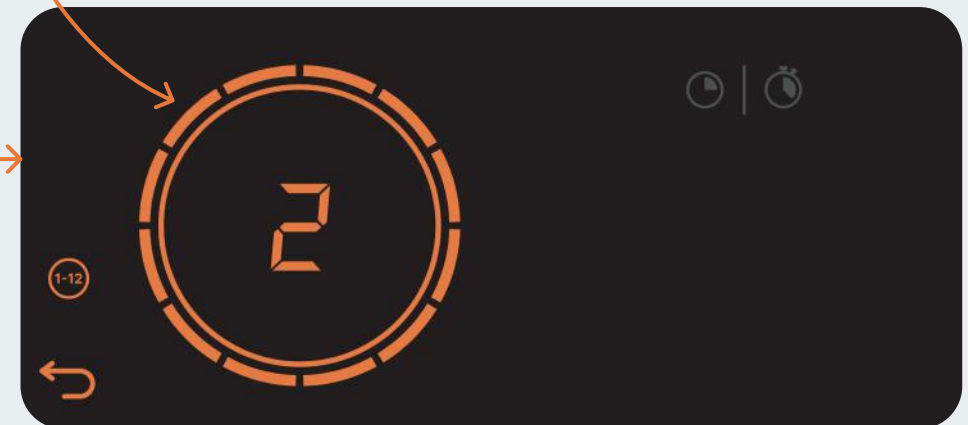


Figure 66 : Hob Interface - Power Level Mode - Level set

General Functions

Timer

The timer function has been redesigned incorporating the following improvements:



Grey Progress Bar

A grey progress bar appears inside the orange Iris slider, displaying the amount of time left visually (Figure 67). This bar stays as a whole single line when the time is in minutes. However, if the time exceeds over an hour (eg., when used for slow cooking), the line is segmented, with each segments representing one hour (Figure 69).



H.M. Text only displayed when required (H.M.)

The H.M. (Hours, Minutes) text only comes up when the timer displays the time in hours and minutes, and not when it is in minutes and seconds. This reduces confusion regarding its meaning/use (Figure 69).

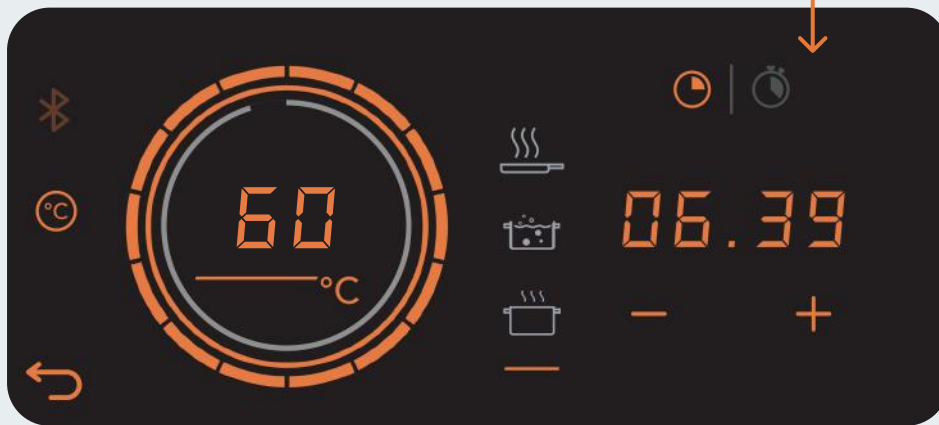
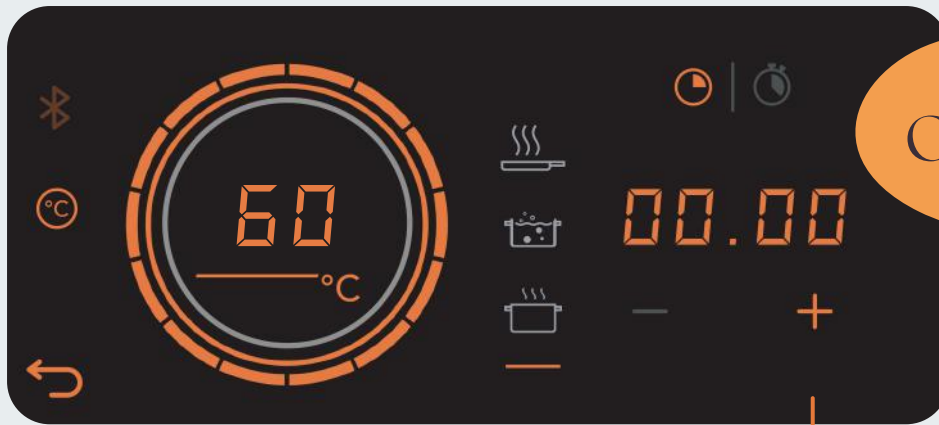


Figure 67 : Hob Interface - Timer function - Setting the timer

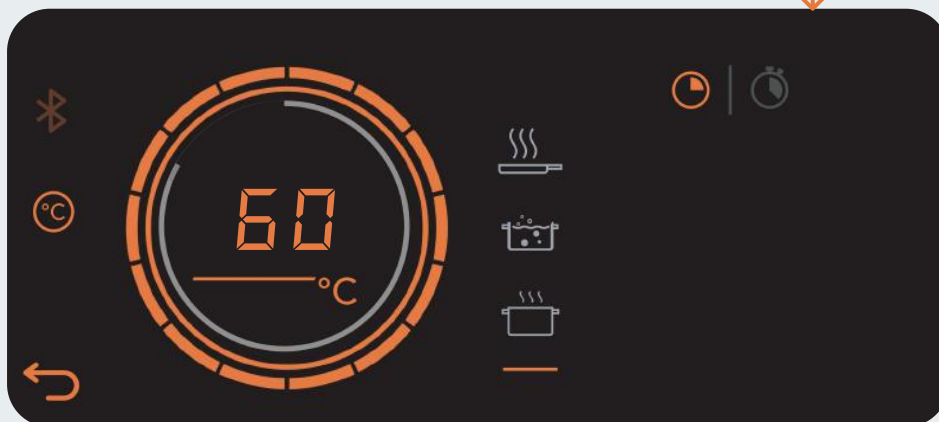


Figure 68 : Hob Interface - Countdown numbers disappearing after 20 seconds

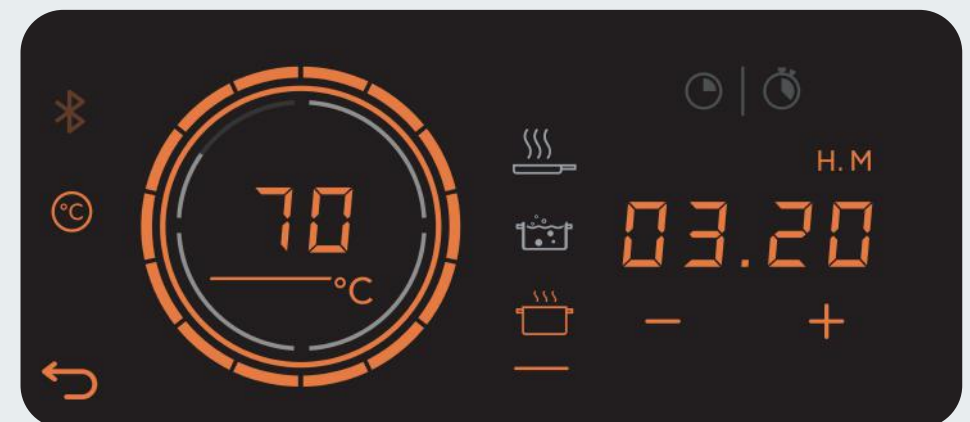


Figure 69 : Hob Interface - Segmented timer for slow cooking

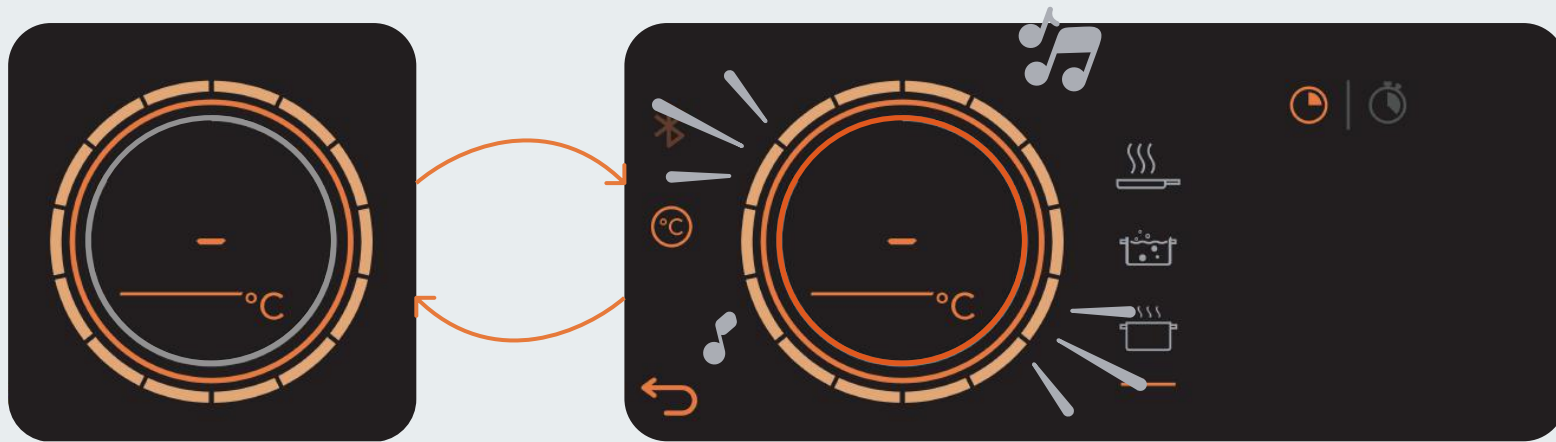


Figure 70 : Hob Interface - Timer function when complete (pleasant sound and blinking of grey bar)

Timer Completion Feedback

On completion of the timer, a pleasant sound plays (same as original hob). However, in order for the user to know which cooking zone's timer has gone off, the grey progress bar turns orange and blinks for 5 seconds. This acts as a visual cue for the user (Figure 70).

Addition to the Settings

As a response to our users who preferred the timer's numbers over the visual grey bar, we added a setting called 'tmr' representing 'Timer' and the ability to switch it on or off. Keeping it on would mean that the numbers of the countdown would be always displayed rather than going off after 20 seconds (Figure 71).



Figure 71 : Hob Interface - Timer setting - Keep the countdown display always on

Bluetooth connection



Figure 72 : Hob Interface - Switching on the Bluetooth function

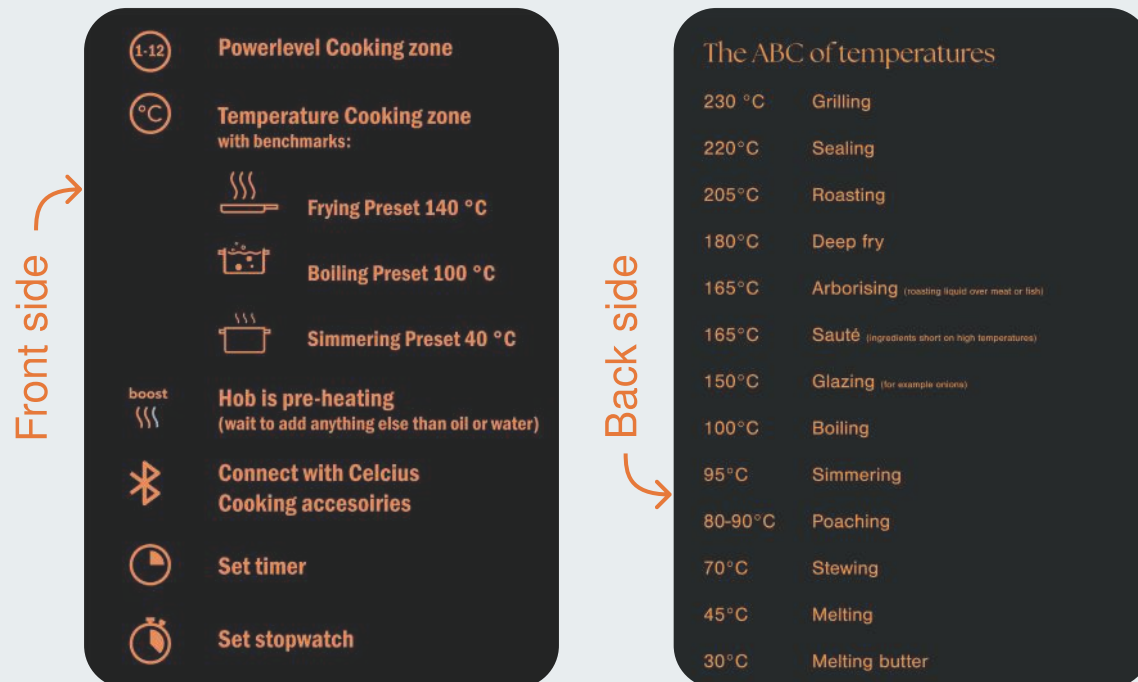


Figure 74 : Visualization of the Quick Glance Menu created for priming/sensitizing the user

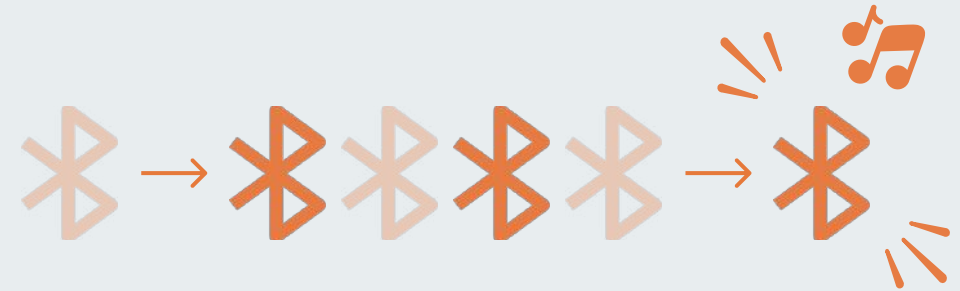


Figure 73 : Blinking of the Bluetooth when turned on

Bluetooth

To make the connection of an accessory to the hob through Bluetooth more explicit, we designed the icon to blink while it's connecting and glow orange when it's connected successfully (Figure 73). This would be accompanied by a sound for additional feedback. More details can be seen in the following sections.

Quick Glance Menu

Using a 'Quick Glance Menu' proved to be successful in our testing, allowing the users to understand and learn the basic functions of the hob before first use. In this quick glance manual, which would be the size of a credit card, a list of the icons along with their meanings are provided. 'The ABC of temperatures' can also be seen on the back side to learn even more about specific temperatures for different kinds of cooking (Figure 74).

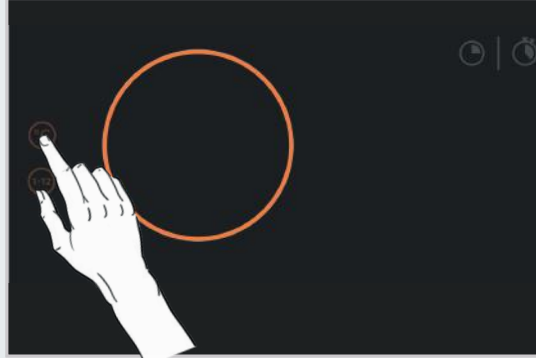
Quick Glance Menu

Final redesign interaction

To show the improved interaction, this storyboard of the redesign interaction is shown. (Figure 60) The phrases in the speech bubbles are quotes of the participants in the final redesign test.



Jack is finally back from a long day of work as a teacher. Finally friday, now he has all the time to explore his new cooking hob. Let's try to make some chicken for him and his wife!



Jack goes to the temperature mode, he wants to cook his chicken at 120 degrees. 'It is so clear which mode I have to use and easy as well!'



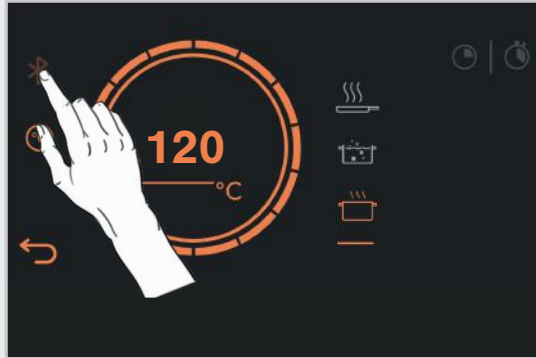
Looking at the quick start guide, Jack knows exactly which default to use. The boiling default goes to 100 degrees, but by swiping around the iris slider it quickly goes to 120 degrees.



Ah it goes to preheat, Jack knows he has to wait patiently with throwing in the chicken. The first time he used the hob, his food got burned because of it. But luckily he learned quickly from his mistake.



Lets set a timer for the chicken. First Jack could only see the visual ring for the timer. Luckily his wife told him he could change this in settings, she read it in the manual. 'I prefer to seeing the exact time always during cooking.'



Jack has been so excited to use the probe. Lets just try to turn it on for only a view seconds. 'I only have to click on the bluetooth buttons.' Connecting... Connecting... It is on! This is so easy to set.



In all the excitement of using the probe, Jack almost forgot to make the sauce. He returns quickly the first screen, to go to the power modes. 'Dinner is almost ready!'



The chicken is perfectly cooked. He tells his wife what he just experienced while cooking. He feels like a real chef now.

Figure 50: The storyboard for our redesigned Atag Cooking hob. Please note that the hands demonstrating the interface interaction are not to scale.

Redesign Hob Accessoires

Before

The Atag Probe and Pan are designed to allow for more precise temperature control. Both accessories are battery-operated and can be connected by **double-tapping the Celsius Cooking Icon on the front** (Figure 75). Once tapped **no product feedback** on the accessories was shown, only this icon **lighted up on the hob** which was often overlooked as one was looking at tapping on the accessory. This connection indication and tapping on a flat interface were **neither intuitive nor effective**.

After

With the integration of celsius within the plus cooking to **one central temperature cooking mode**, we saw the need for additional use cues on the accessories. In the redesign, there will be active feedback on the accessories as well as on the hob (Figure 76). The **blinking bluetooth icon** will communicate the connection status. The active feedback of getting connected and being connected will allow for a more **guided and intuitive interaction**. In Figure 77 the new interaction storyboard is depicted.



Figure 75: The original design of the probe.



Figure 76: The redesigned probe.



Press the bluetooth icon within temperature cooking, and wait for it to blink in search of connection.



Press the bluetooth icon on the probe.



The icon will blink on the probe in search of connection.



The icon on the hob will stop blinking, indicating an active connection.



The icon on the probe has stopped blinking as well and one can enjoy temperature precise cooking!

Conclusion

The final redesign of the Atag Induction cooking hob **involves integrating Celsius cooking with the Plus cooking mode to create a single-temperature cooking mode**. As a result of this integration, the interaction with the Celsius cooking accessories has been updated to provide clearer active Bluetooth feedback. The hob now clearly distinguishes between temperature cooking and power-level cooking, allowing both experienced and novice chefs to use the induction hob more effectively.

With this redesign, users can confidently take control of their cooking in an intuitive manner, leading to increased excitement in managing their cooking behavior. Through this new redesign, **Atag empowers users to unlock the best flavors and feel like a chef at home!**



5 ■ Conclusions & Recommendations

With this chapter, we want to conclude our project. The final design has been explained in the previous chapter and this chapter will include recommendations we make. These recommendations are outside of our design scope, but can be explored to improve the product. This section will showcase recommendations for the hob interface and hob accessories. Finally, we will wrap up this project with a section that expresses our acknowledgements.

Quick Chapter glance:
Hob Recommendations
Hob Interface
User Test recommendations
Acknowledgements



Hob Recommendations

After completing the final redesign, we have pinpointed several areas that could be beneficial in the future. These will be outlined as suggestions for future enhancements, offering direction and ideas for upcoming design iterations.

Hob interface

Error Communication

Error

Water

No Pan

When an error occurs, the hob should display prompt messages or provide voice reminders on the screen. This will help users identify the error without consulting the manual (see Figure 78). Such an approach will make the error easier to understand, enabling users to quickly fix it and maintain their confidence and control.

Figure 78: Error communication

Boost Button



Figure 80: Example of the integration of the boost button.

Given the confusion that the semi-automatic interaction of the Boost function can cause, consider implementing a dedicated touch switch for the Boost feature (Figure 80). Allowing users to engage the function more straightforwardly, enhancing their understanding and ease of use. By isolating this function with a distinct control, users can more confidently utilize the Boost feature without second-guessing their actions.

Personalized Setting



Figure 79: Default temperatures.

It is suggested to introduce a feature that permits users to tailor the user interface. Focuses on the ability to modify the default temperatures based on their usage patterns, Providing a more personalized experience and giving them greater control over the hob. (Figure 79).

Interaction with back button

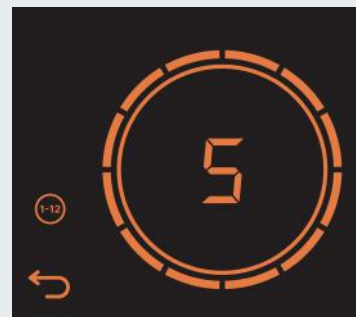


Figure 81: Example of the back button on the interface.

Further investigation is needed into the return function (Figure 81), specifically exploring the capability to go back to previous actions or directly to the menu. Some users preferred for greater flexibility in this feature. Thus, future research could focus on improving the functionality of the stepback feature, possibly allowing a quick tap to go back to the prior action and a long press to access the mode selection menu.

Hob accessories

Celsius Icons on the Hob



Figure 82: Examples of the Celsius cooking accessories.

To make the Celsius cooking function more intuitive and user-friendly, it is advisable to standardize the shape and interaction of icons on both the pan and the probe, and additional accessories in the future (Figure 82). This approach would reduce the learning curve for users, enabling them to quickly adapt to and understand the Celsius functionality on different accessories.

Probe



Figure 83: Allowing stirring with probe.

Taking into account the feedback regarding the inconvenience of using a probe to measure cooking temperatures, particularly when it interferes with stirring and disrupts the cooking process, it could be advantageous to investigate technical enhancements for temperature measurements (Figure 83). This would ensure that it doesn't impede the user's movements in the kitchen but rather supports them.

Usertest Recommendations

During our user test, we encountered **several limitations that have led to recommendations** for future improvements:

- For the initial design refinement Phase, we recommend using **paper cut-outs** resembling raw and cooked food, along with a realistic setup including a pan, spatula, etc. This approach can provide valuable insights into the user experience flow. However, it's important to note that it may **not directly replicate the experience of cooking on a real induction hob**. The use of paper cut-out ingredients may limit the overall enjoyment of cooking, affecting the Hedonic score related to feelings of pleasure. We assume that using paper for cooking inherently impacts the overall enjoyment of the experience.
- We were **limited in using sound** during user testing, However, after completing the tests we realised the importance of sound during the interaction. And therefore, would recommend including sound in further user tests.
- When establishing our personas, we acknowledged the significance of defining them before conducting tests. This ensures that the recruitment of participants is more representative and the data analysis is more objective, unencumbered by the assumptions and perspectives of different participants. It is **advisable to establish personas for both the design and testing Phases** to obtain the most impartial and comprehensive results.

Acknowledgements

The final design has achieved our goal and interaction vision, which we set in the first report. The redesign can be taken further by researching the recommendations for the product.

As a group, we want to thank Arnold Vermeeren and Sonja Paus-Buzink for coordinating this course. We want to thank Martin Verwaal for letting us do our user tests in the applied labs. Also, we want to say thank you to our participants, for helping us research the product and our redesign.

A special thanks go out to our coaches, Renate de Bruin and Lyè Goto, and our clients, Vincent Hofstee, Björn Flinterhoff, and Maureen Naber. Thank you for all the support good conversations and feedback on our work! Thanks to Atag for providing us with the Celsius cooking hob and allowing us to research and work with this product.

We enjoyed this project and learned a broad variety of skills because of it. The given case allowed us to explore our design abilities. As a team, we learned from each other and grew as designers.

Appendix

A: Cognitive Walkthrough
B: Data of pilot test - observer form
C: Test plan
D: Moderator script
E: Observer form
F: Interaction Quality Forms
G: Roles for user tests
H: Prototype
I: PrEmo
J: Data user tests - observer forms
K: Data of user tests - UEQ
L: UEQ comparison
E: What if brainstorm



Appendix A: Cognitive Walkthrough

This table showcases our documentation of the feedforward and feedback cues during the cognitive walkthrough Phase, which took place as a pre-test for the prototype before piloting the main user test.

Task	Feedforward: Will the user know what to do in the next step?	Feedback: If the user does the right thing, will they 1) know that they did the right thing, 2) are making progress towards their goal	Difficult / errors?			
Turning on the hob	Yes, she knew the button	Knew how to place the pan, moderator has to remove the paper and make sound	The waiting time to set to preheat was too short so one had no time to think what to do next and the hob immediately jumped to preheat	we have to say they use the hob in front of them		
Place the pan	Yes, she knew where to place it	Yes, the iris slider appeared				
Set up temperature cooking	Yes, she pressed the right icon	Yes, the celsius showed	She was in boost, it should wait before changing the temperature			
Set up temperature to 30 degrees	No, the sliding function looked like it was jumping. The jumps looked like they worked by itself, when pressed the pan icon for simmering the jump temperature.	No, as the temperature was not at the good temperature now.	The temperature was 10 and not 40, this will now be 45 because of how the figma is set up			
Wait for preheat	Yes, she waited before adding ingredients	No, the preheat went away really fast, missed the sound	Was really quick			
Add butter	Yes, they can add butter now preheat is done	Yes, she with the right temperature could wait for the butter to melt				
Set up temperature cooking 140 degrees	Yes she pressed the jumping to 140	Yes the preheat appeared.	We have to think about changing the temperature			
Wait for preheat	Yes, she wait as the back button and bluetooth disappeared	Yes, she could wait for the icons to turn back on and the pre heat to go away.				
Add oil	Yes, they knew the pan was hot	Yes, the pan was hot				
Add onion, garlic	Yes, they knew the pan was hot	Yes, the onions were frying				
Glaze the onion and garlic	Yes, they fried as expected	Yes, the onions were now glazed				
Set up temperature cooking to 100 degrees	Yes, she pressed the 100	Yes, the 100 degrees showed and jumped to preheat				
Wait for water to boil	Yes, the hob indicated what he was doing	Yes, the preheat disappeared and the water was boiling as it was 100 degrees				
Add the eggs	Yes, the water boiled, temperature set so could add the eggs	Yes, with the eggs they could set the timer now				
Set timer 7 minutes	Yes could easily set up the timer	Yes, the progress bar appeared and the time started running	The timer should go away during the time			
Wait for timer to finish	Yes, they saw the time moving	No, the timer stayed there and was looping in one minute when done	The bar disappears and the timer loops from one minute			
Set up powerlevels	No, the powerlevels were not accessible		The timer stays when on the back button			
Add the hazelnuts	No, added during preheat of frying as they were in the pan	No, the hazelnut were burned				
Roast hazelnut	No, they were burned so there is no next step	-				
Connect the probe	Yes, it was flickering and waiting to connect	No, the flickering went too fast	The flickering should go slower.			
Turning off the hob	Yes, press off and conclude the session	Yes, complete the session				

Appendix B: Data of pilot test - observer form

In this section are the pages of the observer form, where notes from the pilot test with our coach, Renate De Bruin, were taken and used for further refinement of the testing plan for the main user tests.

Observer Form Template

Participant: Renate - pilot
Date: 23-05-'24
Observer name: Iris

Primings
Objective: Explaining the product

Remarks on asked questions
→ has induction
→ uses it everyday

Comments on the explanation of the product (if any)
→ 'okay', 'yes' → understands credit card

Other remarks

Step 1: turn on the hob

Objective: Turn on the hob
Time taken: 30 sec Nr of steps: 2
Look for: turning on hob
Observations:

Interaction flow different steps the user takes
immediately turns it on → no hesitation
places pan

Hesitations/confusions looking at hob, waiting, vocal clues
/

Participant feedback when asked questions
Questions:
• What did you think about the functions?
• Additional thoughts
/

Body language facial expression, whole body
confident

Step 2: Choose mode (temperature) ^{does not need separate step → this time I skipped unknown info}

Objective: user is choosing the temperature mode
Time taken: 8 min 14 sec Nr of steps: _____
Look for:
Observations: use of 'creditcard', icons clear

Interaction flow different steps
places butter → turn on → taps → figma
breaks down → preheat → burns butter →
explanation on why

Hesitations/confusions looking at hob, waiting, vocal clues
'uhm' → trying to find where to turn on.
'oh no' → surprised when trying to set temp
'hey' → excited what happens
'boost, okay, why?' → burning
'how does it boost, automatically?' → 'how to avoid that' butter
!turn of notifications! boost sound, change that it starts at 30.
→ 'this is my usual way'

Participant feedback when asked questions
Questions:
• Are the icons clear to you?
• How did these steps go?
• What did you think of this interaction?

Body language facial expression, whole body
hands crossed → uncertain
→ confused if she used right butter → top icon instead of bottom icon → more intuitive

Step 3: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: 3 min

Nr of steps:

Look for: preheat, presets and sliding for temp. change

Observations:

Interaction flow different steps

still on 45 degrees → adds oil → places onion →
uses slider to go to 140" (?)
↳ did not use preset →

Hesitations/confusions looking at hob, waiting, vocal clues

'I am a bit scared to put onions in but I will do it the way I usually do' → hesitation
'but the pan is already warm, because I melted butter' → more clear to change scenarios

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 4: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: 3 min 53

Nr of steps:

Look for: use of presets, use of 'creditcard', icons clear, preheat

Observations:

Interaction flow different steps

takes different pan → pushes back button → would turn off → places pan → adds water → temp mode → boiling short cut → clicks stopwatch 1st → then timer → + to go to 7 min

Hesitations/confusions looking at hob, waiting, vocal clues

'normally I would turn it off'
'I would choose different hob → pan sizes' ↑ for observer harder to see
'is this turn on, no its already turned on' → more intuitive
'stopwatch, why would you need a stopwatch'
'Ah okay' 'I would push to shut off' → timer
Not wait for eggs to finish → do other first

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 5: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken:

Nr of steps:

Look for: using presets, looking at 'creditcard', setting timer

Observations:

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 6: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____

Nr of steps: _____

Look for: changing modes

Observations: _____

Interaction flow different steps

turn on → uses temp → top icon → waits for
boost → adds hazelnut → roasts

Hesitations/confusions looking at hob, waiting, vocal clues

'i will not put it in. I will wait for some indication that boost is done?' → boost indication went too quick. Does orange mean it's warming up? Other colour or sound. 'Ah I missed the bar' → looks just nice not like giving info
'Is boost / preheat learning curve? Yes if I know why it happens yes. But you told me? ~~Ah~~
Question is: what are you testing if you can't change it'

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

Body language facial expression, whole body

Step 7: Connect probe

Objective: connect probe with bluetooth

Time taken: _____

Nr of steps: _____

Look for: understanding of icon, connection

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

Body language facial expression, whole body

Step 8: fill in UEQ

UEQ feedback

Intuitive aspects

'Credit card not needed much → clear icons'

Confusing aspects

Little obstruction → boost. If you know it's okay. More feedback is nice

Easy

Middle of inefficient - efficient

Not too much info

Interesting

Inventive

Leading edge

Improvement suggestions

Additional notes

Step 9: fill in PreMo

PreMo feedback

Intuitive aspects

Onions: little bit surprised → did it in routine way, simple task annoying when goes wrong. R 3

Egg : 5, went well

Hazelnut: insecure whether I did it correct, could be also because of what we said. nr. 4

Overall:

! Hard to remember!

Confusing aspects

Improvement suggestions

Additional notes

Appendix C: Test plan

In this section, we have laid out our detailed test plan for the final user tests which include testable targets, scenarios/simulated context of use, data collection methods, analysis plan, roles and procedure.

Test Plan:

Purpose

Goal: Determine to what extent our set design goal and vision have been realized and whether we have reached our design criteria and testable targets.

Design goal: We aim to redesign the Atag induction Celsius Cooking Hob and equip regular cooks with the confidence to let the Atag cooking plate empower them through their culinary experimentation in an intuitive manner. Our goal is to make them feel in control and excited about the cooking process, with the integration of new technology that Atag unlocks in their daily lives.

Scope: We are focusing on the Temperature cooking mode (this is now one mode with PLUS and Celsius integrated)

Out of scope: Interface changes of the settings and lock function.

Research questions & Testable targets:

Testable targets:

- The timer function
- The setting of the temperature
- The guidance of the pre-heat
- Changing between modes (Temperature and power)
- Changing between temperatures (after preheating)

Research questions:

Intuitive:

Is operating the redesigned Atag Cooking hob intuitive? Did it respond as expected, and was the interface easy to understand? Was the user flow smooth without much hesitation?

- You feel like the hob responds to the interaction as expected
 - rate from 1-6
- The interface is easy to understand
 - rate from 1-6
- Cooking with the hob is smooth and doesn't give you any hesitations
 - rate from 1-6

In control:

Is the redesigned Atag Cooking hob designed to provide users with a sense of control, making it easy to access functions, manage their time effectively, cook in their preferred style, and use multiple functions simultaneously?

- You could access functions easily
 - rate from 1-6
- You felt in control with your tempo of cooking with the hob
 - rate from 1-6
- The hob allowed you to cook in your way (their usual cooking style)
 - rate from 1-6
- The hob allowed you to switch easily between multiple cooking modes (temperature and power level)
 - rate from 1-6
- The hob allowed you to switch easily between multiple cooking presets (within the temperature mode)
 - rate from 1-6

Guiding:

Does the redesigned Atag cooking hob assist users by providing clear feedback on their actions, allowing them to rely on the hob to guide their cooking process effectively?

- The hob provided clear feedback on your actions
 - rate from 1-6
- You could rely on the hob to perform as you instruct it to
 - rate from 1-6
- The hob steered you in the right directions
 - rate from 1-6

Exciting:

Is the redesigned Atag cooking hob designed to offer an engaging cooking experience, allowing users to prepare food to their liking and encouraging them to explore new functions and unleash their culinary creativity in the kitchen?

- With this hob, you can cook your food the way you like it
 - rate from 1-6
- You see yourself experimenting your cooking style with the hob
 - rate from 1-6
- The hob supports you in learning about new functions
 - rate from 1-6
- You are eager to use the hob for cooking
 - rate from 1-6

Negative	Positive
obstructive	supportive
complicated	easy
inefficient	efficient
confusing	clear
boring	exciting
not interesting	interesting
conventional	inventive
usual	leading edge

Simulated context of use

Home Kitchen: Spices, cutting board, different ingredients in preparing a meal, a plant, oil, kitchen appliances (plates, cutlery, etc.).

- Print spice rack
- Print a home kitchen atmosphere

Bringing for the test:

- The celsius pan
- Ingredients (print)
 - oil
 - onion + garlic, and then fried
 - water, boiling
 - egg (for boiling)
 - green vegetables
 - Butter and melted butter

Data Collected:

- Are people's hands in the way of the Iris slider?
- Are people looking at the quick glance guide?
- Did they understand they were locked during the pre-heat?

Data collection methods

- Observations
 - Make template that coordinate with the research question
 - User performance (what do they perform) and how long do they take to do the tasks.
 - Steps taken by the user individually, to prepare for each task
 - What issues to observe
 - valence (positive or negative)
 - who (user/participant)
 - when (use case/ goal/ tasks / conditions)
 - issue description (effectiveness, efficiency, etc.)
 - sources:
 - product/service; component, property
 - user properties; capability, physical property, expectation
 - symbiotic elements (not made for your mobile phone to watch for example)
 - other people
 - environment
- Interviews
 - User experience; what do they expect and what do they know.
 - Priming
 - Their experience with induction and with cooking etc.
 - UEQ per task
- What instruments?
 - Camera (Phone)
 - Screen Figma on a black phone
 - Charger 2x
 - Forms with set scales (**User**)
 - Script, Credit card with explanation for **User**
 - Script (**Moderator**)
 - Template of data collection for **Observer**
 - Emotion scale (Me, myself and EI) PreMo
 - Cookies for participants

Data to be collected:

- **Task:** What is the participant doing? (And why)
- **Effectiveness:** Were they able to complete the task? On a scale of easy, medium and difficult
- **Errors:** What difficulties did they encounter?
- **Satisfaction:** Description of whether they seem satisfied or not
- **Efficiency:** How long did it take to complete the task?
- **Number of actions:** Number of actions needed to complete each task

7. Fill in the rates and UEQ's
 - Let them fill out the UEQ and emotions + ask for comments/notes/thoughts
 - Filling in the forms

Steps of test (Interviewers)

1. Priming the participants (explaining a bit about the scope, a little about the functions and have a summary manual little)
2. Preparing the 'ingredients'
3. Set up the context of use
4. Encourage them to think out loud.
5. Clean the phone in between really well (clean off the finger)
6. Wrap up with your cookies *yeah*

Participant and location

The Participants: Cooks who have experience with an induction cooker and cooks more than one time in the week.

Location: Greeting in main hall, Prepping inside and testing in applied labs, discussion in booth in main hall.

Data Analyse Plan

Make an observation sheet for each participant which can be put in a set google form to make data comparison easier.

We can analyze the data to find the pain points (weak and strong) of the product.

We will use the UEQ questionnaire to analyze the attitude of the participants towards the product.

Data to be collected: **Task:** What is the participant doing? (And why)

- **Effectiveness:** Were they able to complete the task? On a scale of easy, medium and difficult
- **Errors:** What difficulties did they encounter?
- **Satisfaction:** Description of whether they seem satisfied or not
- **Efficiency:** How long did it take to complete the task?
- **Number of actions:** Number of actions needed to complete each task
- **Overall experience:** Overall satisfaction on a scale of positive, moderate or negative.

- **Overall experience:** Overall satisfaction on a scale of positive, moderate or negative.
- Detailed notes

Roles

- Prep person
- Camera setup (no person)
- **Observer (Sitting a bit away)**
- **Moderator**
- Cleaner (can switch between Moderator and Observer)
- Analysis

Procedure

Steps of test (Participants)

Before the day of testing: Send out a consent form and give a small introduction of the test and the credit card. Make a questionnaire to derive basic data. (Maybe change the old one?)

- What do you do first before using a new appliance?
- REMIND the user to bring the consent form (change the consent form)

Before (on the day of testing): Little Chit chat explaining why to test, small explanation

- **Priming them**
 - Explain to them about the functions
 - Not SHOW the functions but TELL them it exists (can show in the manual but not on the hob)
 - Timer
 - Power mode (state that we focused on the temperature function)
 - Temperature mode
 - Not giving too much info
 - Explain how to think out loud (instructions for the test)
 - How they're not being tested, the appliance is
 - Need to let them use it as they do in real life
 - The 'credit card' is available to use at any time throughout

1. Turn it on
2. Choose mode (→ Temperature)
3. Melt butter at 30 degrees
 - a. *for us to clean out the pan*
4. Add oil and Fry onions at 140 degrees
5. Boil an egg (100 degrees) and set a timer of 7 minutes
6. You need quick roasted hazelnuts so go to power levels to roast them

Appendix D: Moderator script

In this section is the script used by the moderators during the user testing, allowing them to guide the users through the scenarios and as a reference for the questions to be asked to them.

Moderator Script for ATAG Celsius Cooking Hub

Redesign test

Introduction (2 minutes)

Greeting

- Thank you for participating in our redesign test of Atag Celsius Induction hob today. Your feedback is very important in improving the user experience of this redesign. The session will take about half an hour ending with cookies as a thank you, and we'll go through a series of tasks ranging from simple to more advanced functionalities. Feel free to share all your thoughts and questions as we proceed.

Consent form (2 minutes)

- Filled by participant before the test

Priming (6 minutes)

- Before starting, I would like to ask some questions
 - At what time of the day do you usually cook?
 - How familiar are you with induction cooking?
- The Celsius Cooking Hub offers a variety of functions designed to enhance your cooking experience, including adjustable power levels, a timer, temperature cooking, and a probe + celsius pan used for the specific temperature program.
- Today, you'll get to explore these features firsthand.
 - Timer
 - Normal (power levels)
 - Temperature (temperature and different cooking styles)
 - simmering, boiling, frying
 - Probe
 - bluetooth by tapping twice, probe and pan
- Our redesign is especially focussed on the temperature cooking, this is also what the test will mainly focus on. You will get a card with a bit of explanation on all the icons that are used in this mode. See it as your mini manual.

- Explain how they should try thinking out loud during the cooking, say the steps they are taking out loud and what they think out loud.
 - How the appliance is being tested and not them
 - Need to let them use it as they do in real life
 - The mini manual is available to use at anytime throughout
-

Starting the Test:

TIPS:

- Stand by and just observe what the participant does. **Try not to step in the testing process.**

SCENARIOS

- Turn the hob on and place the pan
 - **SCENARIO:** You are enjoying a weekend day and feel like exploring your new induction cooking plate, you are ready to explore the temperature mode. Let's turn the hob on. And go to the temperature cooking
- Choose mode (→ Temperature)
 - Questions:
 - Are the icons clear to you?
 - How did these steps go?
 - What did you think of this interaction?
- Melt butter at 30 degrees
 - **SCENARIO:** Now that you are in the temperature mode, let's start melting some butter, for a butter sauce, the temperature needs to be on 30 degrees
 - Questions:
 - How did these steps go?
 - What did you think of this interaction?
 - *for us to clean out the pan*
- Add oil and Fry onions at 140 degrees
 - **SCENARIO:** The butter sauce is finished. We cleaned out the pan so you can glaze some onions and garlic. Let's add some oil and fry some onions at 140 degrees within the temperature mode.
 - Questions:
 - How did these steps go?
 - What did you think of this interaction?
 - *For us, switch plans*
- Boil an egg (100 degrees) and set a timer of 7 minutes
 - **SCENARIO:** ahhh, these onions are looking good, let's boil an egg! Set the temperature to 100 degrees and set a timer of 7 minutes.

- Questions:
 - How did these steps go?
 - What did you think of this interaction?
 - *For us, remove the pan*
- You need quick roasted hazelnuts so go to power levels to roast them
 - **SCENARIO:** Almost forgot to roast the hazelnuts, oh no, quick let's go back to the power levels and roast the nuts.
 - Questions:
 - How did these steps go?
 - What did you think of this interaction?
 - What did you think of the icons?
- Connect the probe
 - To finish of the day you want to explore the connection of the probe, so you know how to set it up next time
 - Questions:
 - How did these steps go?
 - What did you think of the connection cue?
 - What did you think of the interaction and icons?
- Finished: **SCENARIO:** You are done! You have a delicious meal.
- Fill in the rates and UEQ's
 - Let them fill out the UEQ and emotions + ask for comments/notes/thoughts
 - Filling in the forms
- ---

 Session Closure: Confirm that all participant questions have been addressed, provide any agreed-upon compensation, and thank the participant again for their valuable input.
- Guidance Out: Ensure the participant knows how to exit the building, and express appreciation for their time once more.

Appendix E: Observer form

In this section is the blank template used by the observers during the testing for note taking.

Observer Form Template

Participant: _____

Date: _____

Observer name: _____

Primings

Objective: Explaining the product

Remarks on asked questions

Comments on the explanation of the product (if any)

Other remarks

Step 1: Choose mode (temperature)

Objective: user is choosing the temperature mode

Time taken: _____ **Nr of steps:** _____

Look for:

Observations: use of 'creditcard', icons clear

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- Are the icons clear to you?
- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 2: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: _____ **Nr of steps:** _____

Look for: preheat, presets and sliding for temp. change

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 3: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: _____ **Nr of steps:** _____

Look for: use of presets. use of 'creditcard', icons clear, preheat

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 4: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken: _____ **Nr of steps:** _____

Look for: using presets, looking at 'creditcard', setting timer

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 5: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____ **Nr of steps:** _____

Look for: changing modes

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

Body language facial expression, whole body

Step 6: Connect probe

Objective: connect probe with bluetooth

Time taken: _____

Nr of steps: _____

Look for: understanding of icon, connection

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

Body language facial expression, whole body

Step 7: fill in UEQ

UEQ feedback _____

Intuitive aspects

Confusing aspects

Improvement suggestions

Additional notes

Step 8: fill in PreMo

PreMo feedback

Intuitive aspects

Confusing aspects

Improvement suggestions

Additional notes

Appendix F: Interaction Quality Forms

Here are the interaction quality forms created for the users to rate their experience with the redesign at the end of their tests.

Intuitive

This section tests whether the interaction with the hob felt intuitive.

1. You feel like the hob responds to the interaction as expected?



2. The interface is easy to understand



3. Cooking with the hob is smooth and doesn't give you any hesitations



In control

This section tests whether the hob let you feel in control.

4. You could access functions easily



5. You felt in control with your tempo of cooking with the hob



6. The hob allowed you to cook in your way (your usual cooking style)



7. The hob allowed you to switch easily between multiple cooking modes (temperature and power level)



8. The hob allowed you to switch easily between multiple cooking presets (within temperature mode)



Guiding

This section tests whether the hob guided you in the cooking process.

9. The hob provided clear feedback on your actions



10. You could rely on the hob to perform as you instruct it to



11. The hob steered you in the right directions



Exciting

This section aims to find out if the interaction was exciting.

12. With this hob, you can cook your food the way you like it.



13. You see yourself experimenting your cooking style with the hob



14. The hob supports you in learning about new functions



15. You are eager to use the hob for cooking



Appendix G: Roles for user tests

Roles for the testing conducted during Phase 3

Moderators - Cato and Lois

The role of the moderator is to guide the participant during the test by explaining the tasks to the participants, along with asking questions or answering doubts when needed.

Note takers/Observers - Harshita, Iris and Qinlin

The note taker/observer observes the participant performing the user test and notes down information that is useful for the research. The observer notes down:

quotes/comments

emotions

changes in body language

number of steps taken per task

time taken per task

All of this is jotted down with pen and paper during the user test.

Roles for each test

Participant 1 - Cato + Harshita

Participant 2 - Cato + Iris

Participant 3 - Lois + Harshita

Participant 4 - Lois + Iris

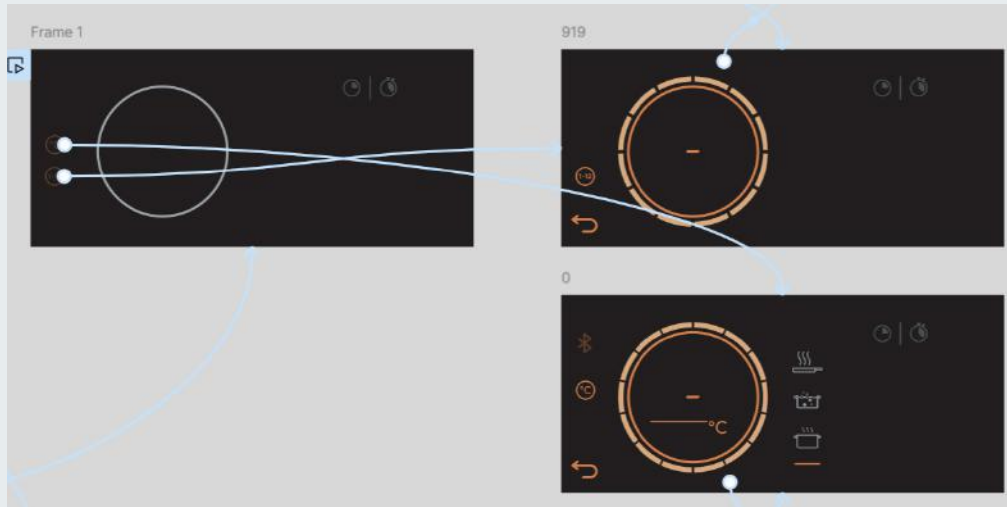
Participant 5 - Cato + Qinlin

The team members who were not testing per participant would be collecting and organizing all the data collected. Overall analysis was done with all team members after the testing Phase was concluded.

Appendix H: Prototype

Following images explains the prototype flow of the final redesign on the Flgma interface.

The start page and the mode menu.

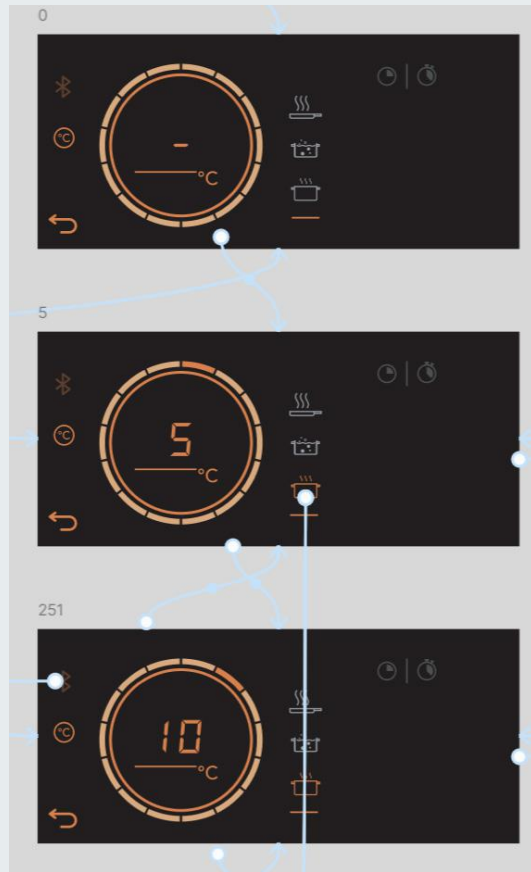


The bluetooth interaction flow is shows as below.

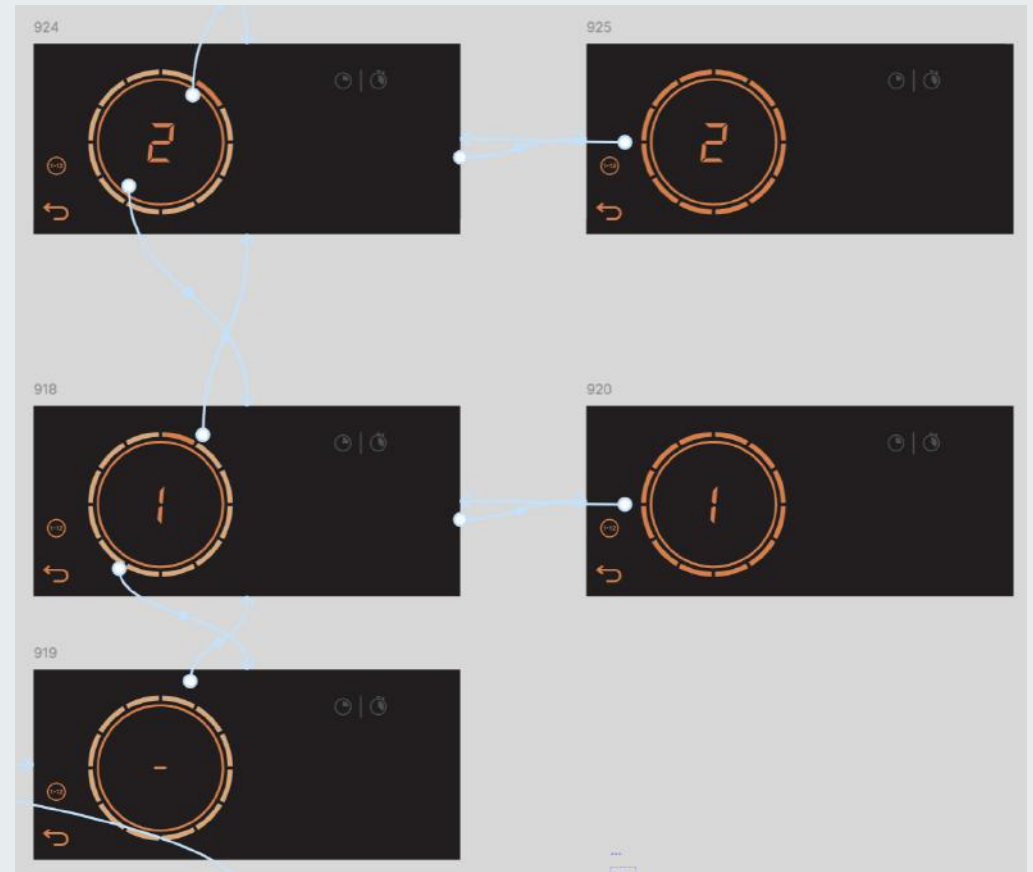


Appendix H: Prototype

Interaction flow in the temperature mode



Interaction flow in the level mode

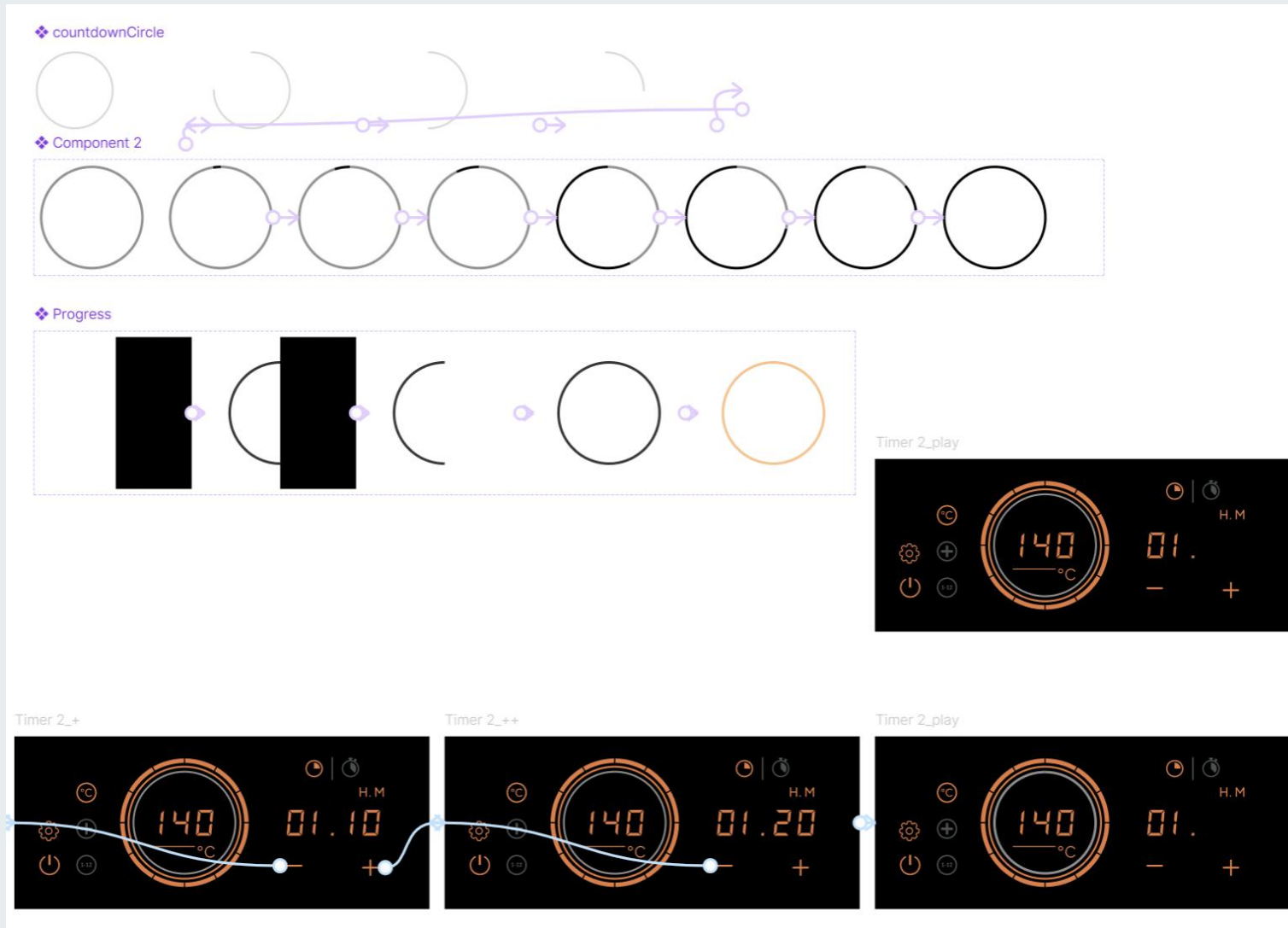


Animation flow of the boosting process



Appendix H: Prototype

Animation flow of the timer function and the iris slider



Appendix I: PrEmo

PrEmo



Introduction

Even though we experience many emotions every day, it is often difficult to find the right words to communicate them. What if we could simply point out emotional expressions to show how we feel?

This idea inspired PrEmo: A pictorial emotion self-report instrument.

The heart of PrEmo is a cartoon character that uses his face, body and voice to express fourteen different emotions. The fourteen cards can be used for measuring emotions and to support people in talking about their feelings.

With the free PrEmo app, you can bring the characters to life. Because the cards do not rely on words, they support cross-cultural research. The characters have been validated in The Netherlands, UK, China, and the USA.

This manual gives you

- instructions for obtaining the application
- definitions of the fourteen PrEmo emotions
- tips and suggestions for using the card set
- colophon with copyright information

PrEmo Emotions

For more information about the emotions, see: www.emotiontypology.com

Pride

The feeling when you possess (or have accomplished) something that exceeds your own expectations, or that others find praiseworthy.



Shame

The feeling when someone knows something bad about you. You believe this to be true, so you also feel bad about yourself.



Admiration

The feeling when you look up to someone who has excellent abilities or impressive accomplishments.



Contempt

The feeling when someone is inferior or unworthy in your eyes; they have done something blameworthy.



Joy

The feeling when something good happens to you. You fulfil a need, achieve something, or make progress towards achieving a goal.



Sadness

The feeling when you have lost something that was important to you, and believe it cannot be undone.



Hope

The feeling when you believe (but are not certain) that something good or desirable may happen in the future.



Fear

The feeling when something can harm you or someone you care about, and you are not sure whether you can do anything about it.



Satisfaction

The feeling when something meets or exceeds your expectations.



Anger

The feeling when someone does something bad that harms or offends you, and you blame that other person.



Desire

The feeling when you anticipate that something will be beneficial once acquired or consumed.



Disgust

The feeling when you find something repulsive and want to avoid having contact with it in any way.



Fascination

The feeling when you encounter something new and interesting that you do not immediately understand.



Boredom

The feeling when there is nothing interesting or engaging to do or experience.



Desmet, P.M.A. (2019). PrEmo card set: Male version. Delft, Delft University of Technology. ISBN: 978-94-6384-076-7.

Appendix J: Data user tests - Observer forms

In this section are the pages of all the observer forms from the five user tests, where notes were taken and used for the analysis that led to the redesign.

Participant 1 - Observer Form

Observer Form Template

Participant: Job
Date: 28
Observer name: HANSHITA

Primings

Objective: Explaining the product

Remarks on asked questions

Comments on the explanation of the product (if any)

Other remarks

Step 1: turn on the hob

Objective: Turn on the hob

Time taken: _____

Nr of steps: _____

Look for: turning on hob

Observations: _____

Interaction flow different steps the user takes

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Step 2: Choose mode (temperature)

Objective: user is choosing the temperature mode

Time taken: _____

Nr of steps: _____

Look for: _____

Observations: use of 'creditcard', icons clear

Interaction flow different steps

*really easy & intuitive to
"because of quick glance manual
you don't need manual"*

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- Are the icons clear to you?
- How did these steps go?
- What did you think of this interaction?

*pretty clear
first time use, you don't know of ea
increment
⑥ new but interesting to see how
it works*

Body language facial expression, whole body

laughed when waiting for butter to melt

Appendix J: Data user tests - Observer forms

Step 3: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: _____ **Nr of steps:** _____

Look for: preheat, presets and sliding for temp. change

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 4: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: _____ **Nr of steps:** _____

Look for: use of presets, use of 'creditcard', icons clear, preheat

Observations: _____

Interaction flow different steps

presses temp. → frying mode → waits for
heating up → puts oil → puts onion + go
(empty pan)

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

similar to the butter
easy → select temp. preset
dear → "don't need a manual"

Body language facial expression, whole body

③ → figured out yourself without manual
(felling proud)

Step 5: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken: _____ **Nr of steps:** _____

Look for: using presets, looking at 'creditcard', setting timer

Observations: _____

Interaction flow different steps

pan on → °C → boil mode →
puts timer ← puts water + eggs ← wait for
pan to heat up

Hesitations/confusions looking at hob, waiting, vocal clues

"Is it minutes?"
taps the timer to show the time left
(tech problem → figure imitation)
Timer shows H.M. but it's actually M.S.
(didn't get immediately)
didn't notice the grey bar

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

③ or ⑤
same with frying onion and
garlic →
figured out without manual
by himself.

Body language facial expression, whole body

timer done → clear
"maybe if time"

multiple timers → multiple hobs →
maybe not
(need blinking?) so clear

Appendix J: Data user tests - Observer forms

Step 6: Hazelnuts

Step 7: Connect probe

Objective: connect probe with bluetooth

Time taken: _____ Nr of steps: _____

Look for: understanding of icon, connection

Observations: _____

Interaction flow different steps

puts hazelnut in pan → sets power level
turns heat off → removes pan & takes it to another heat source (that is off)

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

clear → like other induction hobs
first you figure out how hot
certain

Body language facial expression, whole body

① enjoy cooking

Step 7: Bluetooth

Step 6: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____ Nr of steps: _____

Look for: changing modes

Observations: _____

Interaction flow different steps

°C → sign → ~~connect~~ "maybe it's connecting"
blinking
start

Hesitations/confusions looking at hob, waiting, vocal clues

"no clear connected sign"

no clear indication that it's connected
clear that it's connecting

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

similar to other bluetooth devices
a sound should be nice
also light on °C cooking
not really necessary

Body language facial expression, whole body

"a bit more vague"

bluetooth → ③ or ⑤

probe → "confused face"
↳ can't find on Premo

Step 8: fill in UEQ

UEQ feedback

Intuitive aspects

supportive
easy to use
"not a lot of menus you should go through"
clear, exciting
↳ "first time use, new!"

Confusing aspects

not that inventive
↳ "only integrated"
more tech → leading

Improvement suggestions

Additional notes

Appendix J: Data user tests - Observer forms

Participant 2 - Observer Form

Step 9: fill in PreMo

PreMo feedback

Intuitive aspects

Confusing aspects

Improvement suggestions

Additional notes

Observer Form Template

Participant: Hongbo Liu

Date: 28-05-2024

Observer name: Iris

Primings

Objective: Explaining the product

Remarks on asked questions

cooks 2 portions → for lunch as well
7 PM - 8 PM

not very familiar with induction. But Qintin has explained a bit about it

Comments on the explanation of the product (if any)

seems to understand → 'mmhm', 'okay', 'yeah'

'when do I turn on boost mode'

'oh okay, so I just need to be aware'

'okay, I understand'

Other remarks

Step 1: turn on the hob

Objective: Turn on the hob

Time taken: _____

Nr of steps: _____

Look for: turning on hob

Observations: _____

Interaction flow different steps the user takes

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Appendix J: Data user tests - Observer forms

Step 2: Choose mode (temperature) + butter

Objective: user is choosing the temperature mode

Time taken: 2 min 5 sec

Nr of steps: _____

Look for:

Observations: use of 'creditcard', icons clear

Interaction flow different steps

takes pan → tries to turn on (is already on) → temp. mode → slides to 30° → immediately puts butter → burnt → waits for preheat → places butter again

Hesitations/confusions looking at hob, waiting, vocal clues

'Oh I thought I could put butter'

'If it looks like this idk if its on/off → have slider orange

'Is not very intuitive if its on'

Participant feedback when asked questions

Questions:

- Are the icons clear to you?
- How did these steps go?
- What did you think of this interaction?

Promo 6 curious
4 hopeful that it works
also can be shocked → if I burn butter.

Body language facial expression, whole body

Step 3: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: _____

Nr of steps: _____

Look for: preheat, presets and sliding for temp. change

Observations:

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 4: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: 7 min 5 sec

Nr of steps: _____

Look for: use of presets, use of 'creditcard', icons clear, preheat

Observations:

Interaction flow different steps

clicks temp. mode → sets to 140 → adds oil → waits for preheat → add onions → stirs

Hesitations/confusions looking at hob, waiting, vocal clues

'I think I can put oil, but wait for onions'

'I need to check presets. Really nice to be able to go quickly'

'Because of last time I know what I can put in'

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Promo 4 happy it can go to 140°
doesn't realise if he can adjust temp. itself
3 proud that he can do it

Body language facial expression, whole body

Appendix J: Data user tests - Observer forms

Step 5: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken: 9 min 15 sec

Nr of steps: _____

Look for: using presets, looking at 'creditcard', setting timer

Observations: _____

Interaction flow different steps

places pan → temp mode → boil preset → add water → wait for preheat → boiling → set timer for 7 min → adds egg
check on timer: clicks on timer icon
tries bluetooth → connect to watch to see timer

Hesitations/confusions looking at hob, waiting, vocal clues

Doesn't realise what bar is for → would know what its for but will still want to see at all times
Wants to have timer visible always.

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Likes PreMo
I'm happy because easy to use preset
I'm confident because hob guides you → don't have to do all by yourself

Body language facial expression, whole body

Step 6: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____

Nr of steps: _____

Look for: changing modes

Observations: _____

Interaction flow different steps

go to power → adds hazelnuts → sets powerlevel → roasts hazelnut

Hesitations/confusions looking at hob, waiting, vocal clues

'I don't know which level I should use for this'
Is similar to temp mode

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

I'm confident → copy temp. mode moves to this mode

Body language facial expression, whole body

Step 7: Connect probe

Objective: connect probe with bluetooth

Time taken: 2 min 10 sec

Nr of steps: _____

Look for: understanding of icon, connection

Observations: _____

Interaction flow different steps

goes to temp mode → clicks bluetooth

Hesitations/confusions looking at hob, waiting, vocal clues

'So if I click this I think it's connection'
'I think this means it's searching for connection'
Is intuitive because it looks similar to other bluetooth connection things.
'How do I know if its connected to probe?' → if there's multiple things to connect to

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

PreMo
3 → really easy to use

Body language facial expression, whole body

Appendix J: Data user tests - Observer forms

Participant 3 - Observer Form

Step 8: fill in UEQ

UEQ feedback

Intuitive aspects

supportive because guides through cooking proces
sometimes I need intro. But after 1st time I know &
will not ~~guide~~ give me problems
easy → icons clear
efficient → grey circle is redundant
very clear

Confusing aspects

exciting → lots of functions. Make cooking more
slightly interesting → because cooking is not interesti
for me
inventive → all things have been used before
but you don't want too much new thing

Improvement suggestions

Additional notes

Step 9: fill in PreMo

PreMo feedback

Intuitive aspects

Confusing aspects

Improvement suggestions

Additional notes

Observer Form Template

Participant: Teon van Lierhout

Date: 28/5/24

Observer name: Harshita

Primings

Objective: Explaining the product

Remarks on asked questions

usually cooks during → 5:30 + 6:30 pm

Comments on the explanation of the product (if any)

understood it

Other remarks

Appendix J: Data user tests - Observer forms

Step 1: turn on the hob

Objective: Turn on the hob

Time taken: _____

Nr of steps: _____

Look for: turning on hob

Observations: _____

Interaction flow different steps the user takes

Hesitations/confusions looking at hob, waiting, vocal clues

did it easily

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Step 2: Choose mode (temperature)

Objective: user is choosing the temperature mode

Time taken: _____

Nr of steps: _____

Look for: _____

Observations: use of 'creditcard', icons clear

Interaction flow different steps

"I will press the 'C' button - because I think that's the temp. zone"

Hesitations/confusions looking at hob, waiting, vocal clues

none

Participant feedback when asked questions

Questions:

- Are the icons clear to you? yes
- How did these steps go?
- What did you think of this interaction? no

Body language facial expression, whole body

Step 3: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: 50s

Nr of steps: _____

Look for: preheat, presets and sliding for temp. change

Observations: _____

Interaction flow different steps

takes non-melted butter → put it in the pan
sets to 40°C

Hesitations/confusions looking at hob, waiting, vocal clues

"I only have the 40°C option"
"So it's first going to boost, so I will wait. But maybe I shouldn't have put the butter in."

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

first time = 3 options 40, 100, 140
so choose one of the options
"It's clear, I wish the temp. was flashing or something."

Body language facial expression, whole body

eager to learn
didn't feel frustration
let's try new experience

} remove

Appendix J: Data user tests - Observer forms

Step 4: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: _____ **Nr of steps:** _____

Look for: use of presets, use of 'creditcard', icons clear, preheat

Observations: _____

Interaction flow different steps

puts pan (confused with changing it) → puts oil → sets temp.
puts onions

Hesitations/confusions looking at hob, waiting, vocal clues

was confused with the recipe
"I would dice my onions, rather"

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

very intuitive
"I didn't do anything special"
"normal day of cooking"

Body language facial expression, whole body

"I remembered the instructions."

Step 5: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken: _____ **Nr of steps:** _____

Look for: using presets, looking at 'creditcard', setting timer

Observations: _____

Interaction flow different steps

puts water → boiling → puts egg → timer
in pan
puts egg

"It goes away, but I hope it's still on"

Hesitations/confusions looking at hob, waiting, vocal clues

"I would like a start button."
"I don't see any numbers."
Didn't initially notice the grey bar

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

"if there was a ~~timer~~ ^{setting} to see the timer (members) always, I would ~~have~~ ^{prefer} that."
"normal, as expected" ^{except the start of timer}

Body language facial expression, whole body

confused

feedback → any sound would be okay
of timer

Step 6: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____ **Nr of steps:** _____

Look for: changing modes

Observations: _____

Interaction flow different steps

replaces pan → sets temp. → puts level hazelnuts

Hesitations/confusions looking at hob, waiting, vocal clues

"there's no indication that I need to wait"
"I'm not sure if it was"
intuitive → with how high it would be
quick → in hurry → high temp.

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

"I'm glad I wasn't stupid, after I saw it worked"
understood the back sign

Body language facial expression, whole body

Appendix J: Data user tests - Observer forms

Step 7: Connect probe	Step 8: fill in UEQ	Step 9: fill in PreMo
Objective: connect probe with bluetooth Time taken: _____ Look for: understanding of icon, connection Observations: _____ Interaction flow different steps puts probe (blinking) on pan's edge "started ^{blinking} finding + ^{stopped blinking} found something" with nothing around, maybe it's connected to the probe Hesitations/confusions looking at hob, waiting, vocal clues "expect something to happen" Participant feedback when asked questions Questions: • How did these steps go? • What did you think of this interaction? • What did you think of the icons? Body language facial expression, whole body	UEQ feedback _____ Intuitive aspects _____ using it once was intuitive second time will be easier "will fly through" Confusing aspects _____ Improvement suggestions _____ start indication for timer button always on display Additional notes _____	PreMo feedback _____ Intuitive aspects _____ Confusing aspects _____ Improvement suggestions _____ Additional notes _____

Appendix J: Data user tests - Observer forms

Participant 4 - Observer Form

Observer Form Template

Participant: Mart Vloet

Date: 28-05-2024

Observer name: Iris

Primings

Objective: Explaining the product

Remarks on asked questions

used to induction → parents have it
only uses it when he's at his parents

Comments on the explanation of the product (if any)

Other remarks

Step 1: turn on the hob

Objective: Turn on the hob

Time taken: _____

Nr of steps: _____

Look for: turning on hob

Observations:

Interaction flow different steps the user takes

presses power on

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Step 2: Choose mode (temperature) + butter

Objective: user is choosing the temperature mode

Time taken: 2 min 55

Nr of steps: _____

Look for:

Observations: use of 'creditcard', icons clear

Interaction flow different steps

places pan → wants to use level → goes back →
selects temp. mode → goes to 80° → takes butter
(waits for preheat) → places butter

Hesitations/confusions looking at hob, waiting, vocal clues

So 'If I like the system I would press levels'

'Choice for 2 modes is clear'

Boost is easy, clear to use. Clear you have to wait
Similar to parent's hob

Participant feedback when asked questions

Questions:

- Are the icons clear to you?
- How did these steps go?
- What did you think of this interaction?

5 because it's easy to use
confident & exiting

Body language facial expression, whole body

Appendix J: Data user tests - Observer forms

Step 3: Melt butter at 30 degrees

Objective: Melt butter in temperature mode at 30 degrees

Time taken: _____ **Nr of steps:** _____

Look for: preheat, presets and sliding for temp. change

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 4: Add oil and fry onions at 140 degrees

Objective: Add oil, fry onions in temperature mode.

Time taken: 5 min 15 sec **Nr of steps:** _____

Look for: use of presets, use of 'creditcard', icons clear, preheat

Observations: _____

Interaction flow different steps

tries slider again → reaches 140° → waits for
boosting → places onion

Hesitations/confusions looking at hob, waiting, vocal clues

'Oh now I have to tap a lot' → more what I'm used
'Oh now I see the preset' ↓
→ increments of 5 is okay I am aware of
presets, but didn't
realize the temps
Feedback preheat is clear & enough → you're focused
on the interface

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

Step 5: Boil an egg at 100 degrees & set timer

Objective: Use of presets and timer

Time taken: 8 min 5 sec **Nr of steps:** _____

Look for: using presets, looking at 'creditcard', setting timer

Observations: _____

Interaction flow different steps

takes other pan → uses preset → sets timer → adds
eggs

Hesitations/confusions looking at hob, waiting, vocal clues

Immediately uses preset → 'Now I know presets'
'I'm always a bit impatient with water'
Little doubt & what was stopwatch & timer → prefers
Notifies grey bar timer
→ with long time
→ it can go & disappear
Would tap timer to see time again → wants time to stay
& not disappear
↓
have
the timer
go after
30 min/
1 hour

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?

Body language facial expression, whole body

→ Sounds different than usual timer → distinct sound
↳ easy to link to hob

Appendix J: Data user tests - Observer forms

Step 6: Quick roast hazelnuts with power levels

Objective: change to power mode

Time taken: _____

Nr of steps: _____

Look for: changing modes

Observations: _____

Interaction flow different steps

is in temp → takes different pan → first tries tapping on temp. icon → uses go back → goes to power

Hesitations/confusions looking at hob, waiting, vocal clues

Doubts → doesn't realise how to go back

'It's nicer that both modes are always there'

↳ Less steps & less annoying

'Annoying when buttons don't work immediately due to grease, water etc.'

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

12 because it wasn't working instantly after that 5

Body language facial expression, whole body

Step 7: Connect probe

Objective: connect probe with bluetooth

Time taken: 2 min 18 sec

Nr of steps: _____

Look for: understanding of icon, connection

Observations: _____

Interaction flow different steps

looks at it → press on probe → taps bluetooth icon → waits for connection → place it

Hesitations/confusions looking at hob, waiting, vocal clues

'I think it would make a sound when it's connected'

↳ same with earphones.

Icon is clear → usual symbol

Blinking = clear

Participant feedback when asked questions

Questions:

- How did these steps go?
- What did you think of this interaction?
- What did you think of the icons?

5 nice

Body language facial expression, whole body

Step 8: fill in UEQ

UEQ feedback

Intuitive aspects

supportive: all things are clear, not full because of go back button

easy: maybe too easy → thought too difficult → presets

very efficient: see supportive

clear: see supportive

exciting: new → interesting but currently no knowledge on temps

Confusing aspects

interesting: because it's new. Not max because 1st few times exciting, after that becomes normal inventive

leading edge: is very useful when you know it

Improvement suggestions

Additional notes

Appendix J: Data user tests - Observer forms

Step 9: fill in PreMo

PreMo feedback

Intuitive aspects

Confusing aspects

Improvement suggestions

Additional notes

Appendix K: Data of user tests - UEQ

In this section are the pages of all the UEQs that were filled in by the participants, which contributed to our quantitative data set for analysis.

Test 1

How did you feel about using the hob?

obstructive	o o o o o ● o	supportive
complicated	o o o o o ● o	easy
inefficient	o o o o o ● o	efficient
confusing	o o o o o ● o	clear
boring	o o o o o ● o	exciting
not interesting	o o o o o ● o	interesting
conventional	o o o o o ● o	inventive
usual	o o o o o ● o	leading edge

Test 2

obstructive	o o o o o ● o	supportive
complicated	o o o o o ● o	easy
inefficient	o o o o o ● o	efficient
confusing	o o o o o ● o	clear
boring	o o o o o ● o	exciting
not interesting	o o o o o ● o	interesting
conventional	o o o o o ● o	inventive
usual	o o o o o ● o	leading edge

Test 3

How did you feel about using the hob?

obstructive	o o o o o ● o	supportive
complicated	o o o o o ● o	easy
inefficient	o o o o o ● o	efficient
confusing	o o o o o ● o	clear
boring	o o o o o ● o	exciting
not interesting	o o o o o ● o	interesting
conventional	o o o o o ● o	inventive
usual	o o o o o ● o	leading edge

Test 4

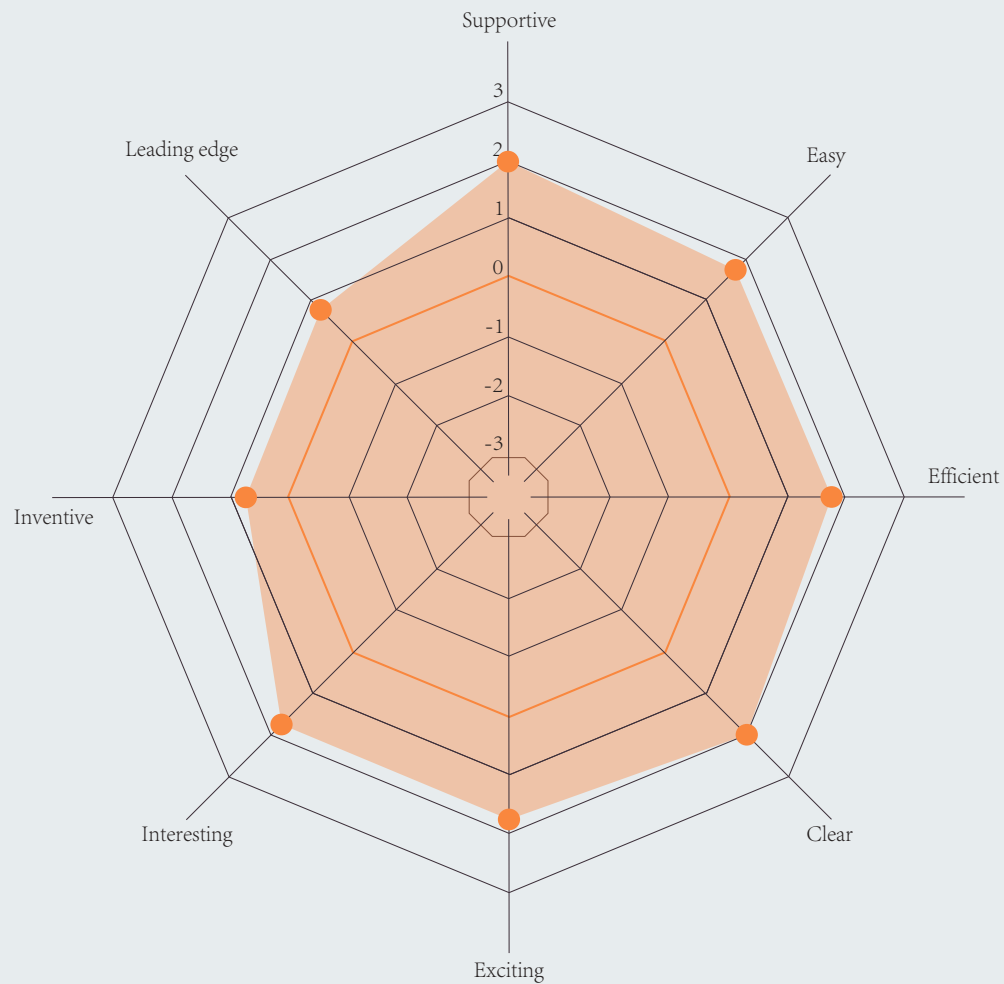
obstructive	o o o o o ● o	supportive
complicated	o o o o o ● o	easy
inefficient	o o o o o ● o	efficient
confusing	o o o o o ● o	clear
boring	o o o o o ● o	exciting
not interesting	o o o o o ● o	interesting
conventional	o o o o o ● o	inventive
usual	o o o o o ● o	leading edge

Test 5

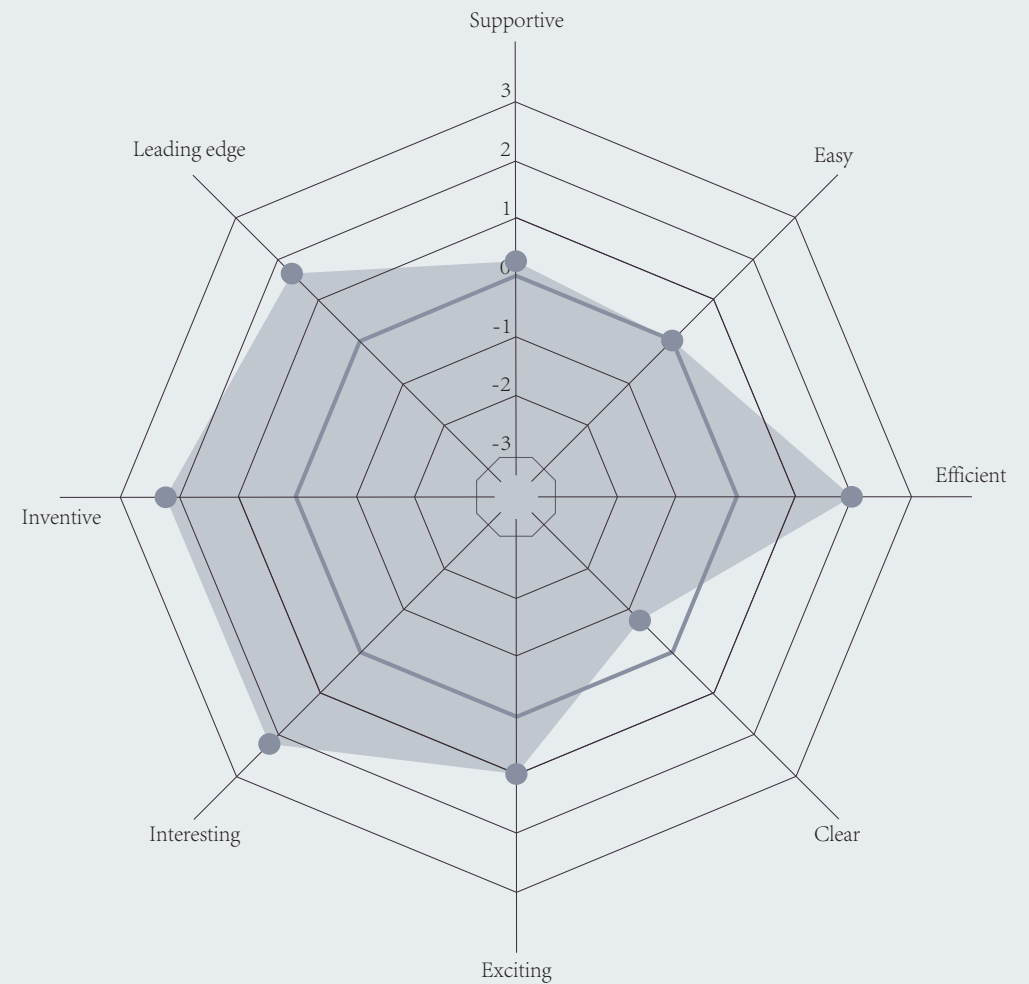
obstructive	o o o o o ● o	supportive
complicated	o o o o o ● o	easy
inefficient	o o o o o ● o	efficient
confusing	o o o o o ● o	clear
boring	o o o o o ● o	exciting
not interesting	o o o o o ● o	interesting
conventional	o o o o o ● o	inventive
usual	o o o o o ● o	leading edge

Appendix L: UEQ comparison

To compare the redesign with the initial design both UEQs of the previous Plus mode, now Temperature mode, are presented. As said in the report these UEQs can not be compared one on one.



UEQ Redesign Phase 3, temperature mode



UEQ initial design Phase 1, Plus mode